# A Minimal-Parameter Framework for Cosmological Dynamics and Galaxy Cluster Mass Discrepancies via Structural Self-Energy Expansion (SSEE)

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**Abstract**

We introduce the Structural Self-Energy Expansion (SSEE), a purely algebraic framework that proposes a connection between cosmological expansion and galaxy-cluster dynamics using only two dimensionless geometric constants — $\pi$ and the golden ratio $\varphi$ — with a minimal effective parameter count and no WIMP, QCD-axion, or SUSY dark-matter particles. The background sector $(w_0, w_a, \Omega_{\rm DE})$ is fully algebraic (zero fitted dimensionless parameters). Phenomenological extensions in the dark-matter sector, proposed in an earlier version of Paper 6, have been *retracted*: the matter sector is a single unpartitioned component $\Omega_m = 0.308881$, and the four register entries those extensions carried (OP-9 through OP-12) close by dissolution — not solved, but no longer questions, since the object they concerned was withdrawn (`OPEN_PROBLEMS.md`; §1.4). From these two axioms, a closed set of seven structural registers is derived algebraically, culminating in the dark-energy equation of state

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi+\pi}{2(\varphi+\pi)} \approx -0.840, \qquad w_a = -\frac{P}{I_g} = -\frac{2\varphi+\pi}{2(\varphi+\pi)} \approx -0.670.$$

Both reduced forms carry the same denominator $2(\varphi+\pi) = 9.519253$, but they descend from *distinct registers*: $w_0$ from $M_v = \varphi+\pi+K_v = 14.278880$ and $w_a$ from $I_g = \pi+P = 9.519253$. It is the lineage, not the reduced form, that fixes each ratio. No fitting procedure is involved: $w_0$ and $w_a$ are uniquely fixed by the geometry of $\varphi$ and $\pi$. The asymptotic structural retention factor $\mathrm{KAL}_0 = \beta+\pi \approx 5.5214$ fixes the cold dark matter density via the forward identity $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, reproducing the observed cosmic dark-to-baryonic matter density ratio. These foundational parameters are tested against data in subsequent papers: a full Bayesian MCMC validation against DESI DR2 [Abdul-Karim et al., 2025], Planck 2018 [Planck Collaboration VI, 2020], and galaxy-cluster mass data ($\Delta\mathrm{BIC} = -7.25$ for the dynamic sector relative to $\Lambda$CDM) is presented in the companion Paper 2 [Almeida Vallejo, 2026], while CMB perturbations and phenomenological extensions are explored in Papers 3–10. The framework is falsifiable: a B-mode polarisation tensor-to-scalar

ratio $r = \varphi^{-10} = 0.008131$ is predicted for LiteBIRD [LiteBIRD Collaboration, 2023] (∼2032).

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# 1 Introduction and Theoretical Foundations

The observed cosmic acceleration [Perlmutter et al., 1999, Riess et al., 1998] requires an explanation beyond the cosmological constant [Weinberg, 1989], motivating a broad landscape of dynamical dark energy proposals [Ratra & Peebles, 1988, Copeland et al., 2006, Weinberg et al., 2013, Clifton et al., 2012]. The SSEE framework operates under the principle of algebraic closure and macroscopic structural self-energy. Rather than adding particle degrees of freedom, the system's identity is defined by a closed set of geometric invariants derived exclusively from $\pi$ and $\varphi$. These fundamental scales dictate the macroscopic behaviour of spacetime:

- $\boldsymbol{\Omega} = \pi + \varphi \approx 4.7596$: Stability Metric.
- $\boldsymbol{\beta} = (\pi + \varphi)/2 \approx 2.3798$: Base Coupling Scalar.
- $\mathbf{KAL_0} = \beta + \pi \approx 5.5214$: Asymptotic Structural Retention. Its law is to *retain*: raising $\mathrm{KAL}_0$ holds more cold matter ($\omega_c = \mathrm{KAL}_0 \omega_b n_s$), lengthens the relaxation time ($\tau_{\mathrm{II}} H_0 = \mathrm{KAL}_0/3\Omega_{\mathrm{DE}}$), lowers the departure ratio ($R_2 = \Omega/\mathrm{KAL}_0 T_r$) and makes the field costlier to move ($X/\mathrm{KAL}_0$) — four uses, one direction. In the hydrodynamic layer of Paper 5 that same retention is what a fluid calls *viscosity*; the name is a manifestation, not a second role. *(Label unified 2026-09-07: earlier versions read "Structural Viscosity", naming the whole constant after its fluid instance — the one use in which* $\mathrm{KAL}_0$ *cancels out of the observable. "Retention" is the name the algebra already carried, in the EFT appendix and in the lineage law of the* $\pi$ *branch.)*
- $\boldsymbol{P} = \Omega + \varphi \approx 6.3777$: Dynamical Evolution Scalar.
- $\boldsymbol{K_v} = \varphi + \pi + \Omega \approx 9.5193$: Structural Constraint Invariant.
- $\boldsymbol{T_r} = 3(\varphi + \beta) \approx 11.9935$: 3D Saturation Horizon.
- $\boldsymbol{M_v} = \varphi + \pi + K_v \approx 14.2789$: Maximal Dimensional Invariant.

Throughout this work = separates a register from its *exact* definition, while $\approx$ introduces only a truncated decimal: $\Omega$ *is* $\pi + \varphi$ with no approximation involved, and it is the four-digit 4.7596 that is approximate, the irrational value having no last digit.

These seven are the structural registers proper. One further register is needed below and is introduced here so that no symbol appears before its definition: $\boldsymbol{I_g} = \pi + P \approx 9.5193$ (exactly $\pi + P$; the decimal is truncated), the *saturation partner* of the dynamical scalar $P$, which is the denominator of $w_a$. It carries the same *value* as $K_v$ (9.519253) while descending from a different construction, and the two are not interchangeable — $K_v$ is a constraint invariant, $I_g$ a saturation partner. Sharing a value is not being the same object, and throughout this work the lineage, not the number, identifies a register (Appendix B).

## 1.1 Predictive Register

A common objection to algebraic models is selection bias: that the constant combinations are chosen post-hoc to reproduce known values. The defence against this is *structural*, not chronological: the constants are drawn from a closed dictionary fixed by construction—no optimisation over formula space is performed. A look-elsewhere scan over the *complete*

closed dictionary of named $(\varphi, \pi)$ constants yields 490 distinct ratios in $(0, 5]$. Each background equation-of-state value is then searched for with a *deliberately generous* tolerance of $\pm 0.0005$ — and exactly *one* of the 490 ratios falls inside it, which moreover matches to machine precision ($|d| < 10^{-16}$). The tolerance is generous and the hit is exact: that is the 1-in-490 specificity, and it persists when the full dictionary, not a hand-picked subset, is scanned (reproducible via `look_elsewhere_full.py`). Table 1 classifies each derivation by epistemic status. We make no claim of temporal priority over public datasets: the quantities compared with DESI DR2 and Planck are postdictions or retrodictions; only those tested against not-yet-released data ($k_{\rm fs}$, $r$, and the pre-registered DESI DR3 test of the fixed $(w_0, w_a)$ point) are genuine predictions. The DR3 entry is a genuine timestamped commitment: the point cannot move, so every tightening of the data is a test it must survive without adjustment. Between DR1 and DR2 the $(w_0, w_a)$ constraint tightened appreciably—from $w_0 = -0.827 \pm 0.063$, $w_a = -0.75^{+0.29}_{-0.25}$ [Adame et al., 2024] to $w_0 = -0.838 \pm 0.055$, $w_a = -0.617 \pm 0.208$ [Abdul-Karim et al., 2025], a reduction of $\approx$33% in the area of the 68% contour (comparing $\sigma_{w_0} \times \sigma_{w_a}$ with the mean of the asymmetric DR1 error on $w_a$: $0.063 \times 0.27 \to 0.055 \times 0.208$)—and the fixed point stayed inside it, at $0.24\sigma$ against the tighter DR2 (Pantheon+-class). Because the point is immobile, the tension scales as the inverse of the error bar, so the pre-registered arithmetic is explicit: a further tightening comparable to the DR1→DR2 step leaves us near 0.4–0.5$\sigma$, while a joint $>3\sigma$ exclusion refutes the algebraic background outright. DR3 (expected 2027) will settle it.

Table 1: Predictive Register: SSEE derivations by epistemic status. "Postdiction": algebraic result fixed by the closed dictionary, then compared to data already public—no claim of temporal priority. "Prediction": fixed now, tested against not-yet-released data. "Retrodiction": derived independently, then compared to an existing measurement. No optimisation over the algebraic formula space was performed.

| Quantity (algebraic formula) | Value | Test / Falsifier | Status |
|---|---|---|---|
| *Background sector (algebraic dictionary)* | | | |
| $w_0 = -T_r/M_v$ | $-0.839950$ | DESI DR2 (2025) | Postdiction |
| $w_a = -(\Omega + \varphi)/I_g$ | $-0.669975$ | DESI DR2 (2025) | Postdiction |
| $(w_0, w_a)$ vs DESI DR3 | same fixed point | DR3 $w_0w_a$CDM (2027); $>3\sigma$ joint excl. falsifies | **Prediction (pre-registered)** |
| $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ | 0.119514 | Planck 2018 | Retrodiction |
| $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2$ | 0.308881 | Planck 2018 | Retrodiction |
| $n_s = 1 - \varphi^{-7}$ | 0.965558 | Planck 2018 | Retrodiction |
| $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen})$ | $67.962\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ | SH0ES input (Post. D); target $3(\varphi + \pi)^2$ | Global background rate |
| *Paper 5 — IS causal perturbation theory* | | | |
| $\gamma_{\rm IS}$ | 0.5504 | RSD surveys ($\gamma_{\Lambda\mathrm{CDM}} = 0.55$) | Retrodiction |
| $f\sigma_8(z{=}0.5)$ | $0.70\sigma$ mean tension | 6 RSD surveys, single-sector (BOSS/eBOSS/6dFGS) | Retrodiction |
| *Paper 6 — growth sector, single component* | | | |
| $S_8$ (KiDS-Legacy, $A_s$ fixed by the CMB) | 0.8273 predicted | measured $0.8265 \pm 0.0176$ ($0.04\sigma$); *no* free cosmological parameter | Prediction |
| earlier: KiDS-1000, $A_s$ fitted (superseded by the row above) | $0.7559 \pm 0.0189$ | $\Lambda$CDM control, same data: $0.7571 \pm 0.0194$ | Retrodiction |
| *($m_\phi$, $k_{\rm fs}$ and the two-sector $S_8$ are retracted; see §1.4.)* | | | |
| *Paper 7 — Canonical EFT, Bellini-Sawicki $\alpha$-functions* | | | |
| $\alpha_T = \alpha_M = \alpha_B = 0$ (minimal coupling) | 0 exact | GW170817 $\lvert\alpha_T\rvert < 10^{-15}$ | Retrodiction |
| $\alpha_K = 3\,\Omega_{\rm DE}(5 - 3w_0)$, $\Omega_{\rm DE} = 0.691119$ | 15.591336 | — (kineticity is not constrained by linear-scale data) | Derived |
| *Future* | | | |
| $r = \varphi^{-10}$ | 0.008131 | LiteBIRD ~2032 | Prediction |

"Retrodiction" denotes results derived independently and then compared against previously published values; no optimisation over the algebraic formula space was performed to match a target value. The distinction between postdictions, retrodictions, and genuine predictions is explicitly noted to maintain transparency. All entries with "Prediction" status are falsifiable by named experiments within the stated time horizon. Where the forecast sensitivity already resolves the predicted value, a measurement excluding it at high significance refutes the framework; where it does not, the entry is marked *below sensitivity* and a non-detection carries no evidential weight either way.

*Note on $\alpha_K$ and $s_K$.* The kineticity of the Paper 7 action is $\alpha_K = 3\,\Omega_{\rm DE}(5 - 3w_0) =$

15.591336, with the density $\Omega_{\rm DE} = 0.691119$. It must not be confused with $s_K = -3w_0(1+w_0) = 3\,s_{\rm DE}\,s_m = 0.403302$, the fractional dilution rate of the dark-energy *pressure* per e-fold: pure equation of state, with no $H$, no density and no $K(X)$ in it (Paper 7, Eq. (sK)). Every use of 0.403302 in this suite — including $f_{\rm scr} = s_K/(3\,\mathrm{MIRA})$ in Paper 9 — is a use of $s_K$.

## 1.2 Axiomatic Principles

SSEE is explicitly constructed as a top-down theoretical ansatz. Its internal symmetries are not phenomenological fits derived bottom-up, but fundamental axiomatic identities that the macroscopic universe is proposed to satisfy. Three physical principles govern the interactions of these geometric invariants:

1. **The Temporal Triad (Cosmological Epochs):** Spacetime evolves through three distinct thermodynamic states — the Origin/Law phase (initial singularity), the Ignition/Processing phase (structure formation), and the Transcendental Saturation phase (dark energy asymptotic domination). The Hubble history $H(a)$ is bounded by:
$$\lim_{a\to 0} H^2(a) \propto \rho_{\rm bar}, \qquad \lim_{a\to\infty} H^2(a) = \frac{8\pi G}{3}\,\rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right). \tag{1}$$

2. **Non-Self-Addition (Evolution by Difference):** Structural evolution proceeds through non-linear geometric scaling rather than linear superposition. All macroscopic observables $\mathcal{O}$ must be strictly functional dependencies of the invariant set and $H_0$; no additional free parameters are permitted:
$$\mathcal{O} = f(\Omega,\, \beta,\, \mathrm{KAL}_0,\, P,\, K_v,\, T_r,\, M_v;\, H_0). \tag{2}$$

3. **Asymmetric Equivalence:** Energy states can possess identical geometric values but divergent dynamic functions. What identifies a register is its construction, not its number: $I_g$ and $K_v$ share the value 9.519253 while remaining distinct registers (Appendix B), and every equality of values in this suite is read as a coincidence to be explained by construction, never as an identity of entities.

These three statements fix the framework's *physical* content. They are not, by themselves, the assumption budget: §1.3 recasts what SSEE takes as given versus what it derives in terms of three register-level postulates (D, S, I), after the 2026 $\omega_m$-direct reframe dissolved the former matter-factor postulate M (OP-8 closed). That postulate accounting — not the count of physical statements above — is the authoritative measure of the model's free choices; Principle 1 above supplies the de Sitter endpoint invoked there by Postulate S.

## 1.3 Scope of the Zero-Parameter Claim

The invariant set $\{\Omega, \beta, \mathrm{KAL}_0, P, K_v, T_r, M_v\}$ and its generators $\varphi, \pi$ are all *dimensionless*. What the framework derives and what it takes as input is made precise by **three postulates in total**, stated here as pre-registered axioms rather than left implicit: **two foundational** — D (dimensional anchor) and S (saturation correspondence), given immediately below — plus **one auxiliary** register-level postulate, I (inflationary register),

given after them. A fourth, the former matter-factor postulate M, is *retired*; it is listed alongside I only to record what it was and why it no longer counts. This is the full assumption budget, and it is recapitulated at the end of the section.

**Postulate D (Dimensional Anchor).** Because the invariant set is dimensionless, SSEE predicts only *dimensionless* quantities: equation-of-state parameters $(w_0, w_a)$, density fractions $(\Omega_{\rm DE}, \Omega_m)$, the EFT functions $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$, the spectral observables $(n_s, r)$, and the structural ratios (MIRA, etc.). These are unit-independent and directly falsifiable. Any *dimensionful* quantity — $H_0$, $\rho_{\rm crit}$, particle masses — requires exactly *one* dimensional input, the anchor, for which we take $H_0$. This is the identical status $H_0$ holds in ΛCDM, where it is one free parameter. The "zero free parameters" claim therefore means precisely: *zero dimensionless fitted parameters*. SSEE carries one dimensional input $(H_0)$ and zero dimensionless ones; ΛCDM carries one dimensional input $(H_0)$ and five dimensionless ones $(\Omega_m, \Omega_b, n_s, A_s, \tau)$. This holds for the *background* sector; at the CMB-likelihood level SSEE retains exactly two dimensionless fitted parameters — the primordial amplitude $A_s$ and the optical depth $\tau$, which no algebraic relation yet fixes — so the honest comparison there is $k = 2$ against ΛCDM's $k = 6$, as set out in the effective-parameter accounting at the end of this section. The dimensional input is *measured*, not algebraic: it is the local expansion rate $H_{\rm SH0ES} = 73.04 \pm 1.04$ km s$^{-1}$ Mpc$^{-1}$, the only directly measured value of $H_0$. The global rate is its *output*, $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962142 \pm 0.968$ km s$^{-1}$ Mpc$^{-1}$, with the screening fraction $f_{\rm screen} = 0.069522$ derived beforehand (Papers 9–10). The pure number $3(\varphi + \pi)^2 = 67.962$ carries no units and is *not* an input of Postulate D (§D): it is the dimensionless *target* the output is compared against, and the two agree to $4.2 \times 10^{-6}$ km s$^{-1}$ Mpc$^{-1}$ (a relative $6 \times 10^{-8}$), far below the precision of the SH0ES input. Deriving the absolute scale of $H_0$ from a fundamental scale is the cosmological-constant problem — the ratio $H_0/M_{\rm Pl} \sim 10^{-61}$ is not bridged by any closed algebraic form — and lies beyond SSEE as it does beyond every current framework.

**Postulate S (Saturation Correspondence).** Define the *saturation fraction*

$$s \;\equiv\; \frac{T_r}{M_v} \;\in\; (0, 1). \tag{3}$$

The dark sector is a one-parameter family in $s$, with the density fraction and the equation of state both fixed by it:

$$\Omega_{\rm DE} = s, \qquad w_0 = -s. \tag{4}$$

The two endpoints are physically forced and require no assumption:

- $s \to 0$: $(\Omega_{\rm DE}, w_0) \to (0, 0)$ — the Einstein–de Sitter (matter-only) limit;
- $s \to 1$: $(\Omega_{\rm DE}, w_0) \to (1, -1)$ — the exact de Sitter limit, already imposed by Principle 1.

An immediate consequence is the much-used identity

$$1 + w_0 \;=\; 1 - s \;=\; s_m, \tag{5}$$

which is therefore *not* an independent assumption but the content of Postulate S. We state explicitly what remains a postulate rather than a theorem: the endpoints fix $(0, 0)$ and

$(1, -1)$, but a nonlinear assignment $w_0 = -s^n$ $(n \neq 1)$ would meet the same endpoints. Linearity $(n = 1)$ is the minimal, lowest-order correspondence and is adopted as such; deriving $n = 1$ from a dynamical attractor of the Israel–Stewart sector is identified as open theoretical work.

**Auxiliary structural postulates beyond D and S.** A careful audit of the derivation chain (this paper plus the companion Papers 3–10) reveals that beyond Postulates D and S, the framework rests on *one* additional auxiliary postulate that does not follow automatically from D+S and is introduced as an algebraic-register declaration. A second one, Postulate M, was carried by earlier versions and is now retired; we keep it listed below, marked as such, so that the reduction of the assumption budget is auditable rather than silent:

- **Postulate M (matter factor) — retired.** Earlier versions postulated a factor mapping the dynamical-sector matter density to the CMB-sector one. This postulate is now *dissolved*: there is no matter factor. The CMB-sector matter density is the standard physical observable $\omega_m = \Omega_{m,\mathrm{CMB}} h^2 = \omega_b + \omega_c + \omega_\nu$, with each piece an algebraic register quantity, $\omega_b = (\pi - \varphi)/(3\Omega^2) = 0.022418$ (OP-1), $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.119514$ (a forward identity, matching Planck at $0.40\sigma$), and $\omega_\nu = \Sigma m_\nu / 93.14$. The CMB matter fraction is then *derived*, $\Omega_{m,\mathrm{CMB}} = \omega_m / h^2 = 0.308881$ at the global rate $H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}}) \simeq 67.962\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (no matter factor; OP-8 dissolved; the former two-$\Omega_m$ criterion is withdrawn, §1.4). The equation-of-state-sector identity $s_m = 1 + w_0 = 0.160050$ is not a density. This **closes OP-8**: there is no factor whose dynamical mechanism must be derived. The structural quantity $\mathrm{MIRA} = (3\varphi + \pi)/4 = \mathrm{AURA}/2 \simeq 1.9989$ persists in the register *only* as the screening amplitude in $f_{\mathrm{screen}}^{\mathrm{IR}} = s_K^{\mathrm{IR}} / (3\,\mathrm{MIRA}) = (\pi - \varphi)/\Omega^2$ (Paper 9), consistent with its name (Mirror Ratio). The former geometric and MIRA forms $\simeq 0.32$ are retired as the matter factor. The honest residue is that $\Omega_{m,\mathrm{CMB}}$ now rests on $\omega_b$ (OP-1) and the forward identity for $\omega_c$ — pieces the register already contained, not a new adjustable knob.
- **Postulate I (Inflationary register).** The inflationary sector is an $\alpha$-attractor with $\alpha = \varphi^4/3$ and the horizon-crossing e-fold number has the algebraic form $N_* = 2\varphi^n$ for some integer $n$. What this postulate fixes is the *ansatz form* ($\alpha$-attractor class, the specific algebraic value of $\alpha$, the structural form $2\varphi^n$), not the integer $n$ itself. The value $n = 7$ is a corollary, not an assumption: among integer exponents, $N_* = 2\varphi^n$ falls inside the canonical inflation window $[50, 60]$ uniquely for $n = 7$ ($2\varphi^6 = 35.888544$, $2\varphi^7 = 58.068884$, $2\varphi^8 = 93.957428$; the first falls below the window, the third above it). Equivalently, the exponent is selected by an observational constraint external to SSEE (the standard inflation duration), not by an internal numerological choice. From these inputs follow the standard $\alpha$-attractor relations $n_s = 1 - 2/N_* = 1 - \varphi^{-7}$ (matches Planck 2018 at $0.16\sigma$) and $r = 12\alpha/N_*^2 = \varphi^{-10}$ (LiteBIRD prediction). Neither the $\alpha$-attractor class, the value $\alpha = \varphi^4/3$, nor the form $N_* = 2\varphi^n$ is derived from D+S; together they constitute Postulate I.

The honest count is therefore **two foundational postulates (D, S) plus one auxiliary register-level postulate (I)** — three total, all dimensionless. Postulate M is retired: with the matter factor dissolved, the CMB matter density is a derived register quantity ($\omega_m = \omega_b + \omega_c + \omega_\nu$), not a postulate, and OP-8 closes. The integer $n = 7$ in Postulate I

is also not part of the assumption budget — it is a corollary of the ansatz form combined with the canonical inflation window. These define the full assumption budget of SSEE. The dark-matter entries that used to appear here (OP-9, OP-11) close by dissolution along with the retracted $\varphi$-DM sector, so the budget is now smaller, not larger. (OP-14 is likewise closed — see the neutrino input below.)

**Honest accounting of effective parameters.** The zero-parameter status established above applies strictly to the *background* sector ($w_0, w_a, \Omega_{\rm DE}, \Omega_m$) and to the dimensionless EFT functions of Paper 7. Phenomenological extensions introduce a small number of effective parameters that, while assigned algebraic values in the current version, do not yet possess a derivation from first principles and are tracked as open problems. One entry has left this list: the CMB matter density is no longer an effective parameter — with Postulate M retired, $\Omega_{m,\rm CMB} = \omega_m/h^2$ is derived from the algebraic $\omega_b$ (OP-1) and the forward identity $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ (OP-8 closed). What remains is:

- *Retracted (see §1.4).* The $\varphi$-DM mass of Paper 6, and with it OP-9, no longer exist as open questions: the particle was withdrawn, so there is no multiplier whose UV origin needs deriving. Its neutrino input survives on its own: $\Sigma m_\nu^{\rm active} = 0.0685\,\mathrm{eV}$ now derives from the clean ratio $\mathcal{R}_2 = \Omega/(\mathrm{KAL}\cdot\mathrm{TRIAL})$, removing the former *ad hoc* integer offset within the structural formula for $\Sigma m_\nu$ (full dimensional expression in Table 5; $\mathcal{R}_2$ is one factor of it, not the total), closing OP-14.
- The non-minimal coupling $\xi$ of $\varphi$-DM was previously tracked as a free parameter; with the $\varphi$-DM sector withdrawn there is no such coupling left to fix, and OP-11 closed by dissolution along with it (2026-08-01).

In CMB-likelihood terms this is a $k = 2$ fit: of ΛCDM's six fitted parameters $(\Omega_m, \Omega_b, n_s, A_s, \tau, H_0)$, SSEE fixes four algebraically — $\Omega_b$ via $\omega_b$ (OP-1), $\Omega_m$ via $\omega_m = \omega_b + \omega_c + \omega_\nu$ with the forward identity $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ (the $h$ in $\Omega_m = \omega_m/h^2$ being the anchor itself), $n_s = 1 - \varphi^{-7}$, and the global expansion rate $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen})$ (SH0ES input, derived $f_{\rm screen}$, Papers 9–10) — leaving exactly two free, the primordial amplitude $A_s$ and the optical depth $\tau$. This $k = 2$ vs $k = 6$ count is the one used in the Paper 3 Bayesian model comparison. The two dark-sector ansätze formerly listed above ($m_\phi$; $\xi$) are retracted with the $\varphi$-DM sector and were never counted among the fit parameters; their register entries (OP-9, OP-11) close by dissolution. See `OPEN_PROBLEMS.md` for the dependency chain.

## 1.4 One matter density, and one equation-of-state parameter

Earlier versions of this suite presented a "Two-$\Omega_m$ criterion": the claim that two distinct matter densities, $\Omega_{m,\rm dyn} = 0.160050$ and $\Omega_{m,\rm CMB} = 0.308881$, each entered different observables, with the selection rule given by a free-streaming scale $k_{\rm fs}$ derived from the $\varphi$-DM particle of Paper 6. **That criterion is withdrawn.** Two independent failures, either one sufficient:

1. $\Omega_{m,\rm dyn}$ **was never a density.** It is defined as $1 + w_0$, i.e. minus the dark-energy equation of state — an algebraic identity of the background sector's $w_0 = -T_R/M_V$. It is dimensionless, as a density fraction also is, so arithmetic involving it is always well formed; but an equation-of-state parameter enters the Friedmann equations through the pressure term, never as a matter density. Calling it $\Omega_{m,\rm dyn}$ named it into a category it never belonged to.

2. **The selection rule no longer exists.** $k_{\rm fs}$ was derived from $m_\phi$, and both are retracted in Paper 6 together with the second sector: the subtraction $\Omega_{\phi\rm DM} = \Omega_{m,\rm CMB} - \Omega_{m,\rm dyn}$ that defined the particle's density mixed a measured density with an equation-of-state number, so the particle had nothing to be made of.

The corrected statement is simpler, and has one fewer moving part:

> **There is one matter density,** $\Omega_m = \omega_m/h^2 = 0.308881$, entering every observable that requires a matter density — background geometry $E(z)$, the sound horizon $r_d$, the CMB acoustic scale, the Poisson source of linear growth, and lensing alike. **There is separately one equation-of-state parameter,** $w_0 = -0.839950$, entering wherever an equation of state enters, and nowhere else.

Two consequences are worth stating explicitly, because they run in opposite directions and honesty requires both.

**What is lost.** The $\varphi$-DM sector, the particle $m_\phi$, the free-streaming scale $k_{\rm fs}$, and the pre-registered $P(k)$-suppression prediction that depended on them. That prediction was genuinely falsifiable and is genuinely gone — not falsified by data, but withdrawn because the construction generating it was not physical. The surviving pre-registered falsifiers of this suite are the tensor-to-scalar ratio $r = \varphi^{-10} = 0.008131$ (LiteBIRD, CMB-S4) and the fixed $(w_0, w_a)$ point against successive DESI releases.

**What is gained.** One postulate fewer, and four register entries closed by dissolution rather than by resolution: OP-9 (UV origin of the mass multiplier), OP-10 (unification of $\varphi$ and $\chi$ into a single field), OP-11 (the free non-minimal coupling $\xi$), and OP-12. None of them was solved; all four ceased to be questions once the object they were about was withdrawn. The suite is also freed of a documented internal inconsistency: a quantity explicitly annotated in the source as "not a matter density" was being summed with one.

**A note on how this was found.** The error was not proposing a two-sector construction (retired 2026-08-01) — that is ordinary model-building, and it was falsifiable. The error was accepting the published $S_8$ as a raw target to fit without first asking whether its data-reduction pipeline already assumed a ΛCDM background. It did. A published number carrying an error bar can still be the output of fitting a fiducial template to the actual measurement; before treating one as a target, the question "raw observable, or model-conditioned summary?" has to be asked explicitly. Paper 6 develops this point and re-does both growth measurements on raw data products.

## 1.5 The ssee Effective Action

Instead of an arbitrary cosmological constant Λ, the vacuum energy density is fixed by the algebraic ratio $T_r/M_v$, leading to the background-level effective action:

$$S^{\rm bg}_{\rm SSEE} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} - \rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right) + \mathcal{L}_{\rm matter}\right]. \tag{6}$$

This action encodes the background Friedmann dynamics only: the dark-energy density is constant at the background level and set by the algebraic ratio $T_r/M_v$. It does not propagate scalar perturbations; the perturbation-level dynamics require a scalar-field extension, developed in the EFT embedding of Appendix A (Eq. 7). Within the Horndeski class of scalar-tensor theories [Horndeski, 1974, Gubitosi et al., 2013, Bellini & Sawicki, 2014], a minimally coupled k-essence scalar [Armendariz-Picon et al., 2001] with the two-term action $K(X) = c_1X + c_2X^2$ and no potential reproduces $w_0$ algebraically while remaining ghost-free and gradient-stable: it is a ghost condensate with adiabatic $c_s^2 = 0.021284$ (Paper 7).

**Falsifiability.** The framework is strictly falsifiable. It would be refuted by: (1) direct detection of a collisionless dark-matter particle accounting for the missing mass; (2) definitive cosmological constraints proving $w_a = 0$; or (3) high-significance confirmation of a normal neutrino hierarchy with $\sum m_\nu < 0.06\,\mathrm{eV}$.

# 2 Effective Field Theory Embedding

**Scope and limitations of this section.** The background dynamics presented here use the *Eckart first-order* approximation [Eckart, 1940], which is sufficient to derive $w_0$, $w_a$, and the viscosity coupling $\mathrm{KAL}_0$ on superhorizon scales. The Eckart formulation is known to violate causality in the relativistic perturbation sector [Müller, 1967, Hiscock & Lindblom, 1983, 1985]; for perturbation-level predictions (B-mode forecasts, sound-speed constraints) the Israel–Stewart (IS) formulation is required and is developed in Paper 5. The Eckart and IS frameworks agree at leading order in the slow-roll expansion, so all background results below are unchanged by the IS extension.

**Sound speed caveat (EFT embedding).** For a pure power-law k-essence Lagrangian $P(X) = A\,X^n$ the adiabatic sound speed satisfies $c_s^2 = 1/(2n-1)$. With $n \approx -0.095$ (derived below) this gives $c_s^2 \approx -0.84 < 0$, indicating gradient instability in the *power-law IR approximation*. This instability is an artefact of the $P(X) \propto X^n$ ansatz and is resolved in Paper 7 by replacing the power law with the two-term k-essence action [Armendariz-Picon et al., 2001] $K(X) = c_1X + c_2X^2$ with minimal coupling and no potential: a ghost condensate ($c_1 < 0$, $c_2 > 0$) that reproduces $w_\phi = w_0$ through $w_\phi = K/(2XK_X - K)$ and has $c_s^2 = 0.021284 > 0$, free of ghost and gradient instabilities. (The functional $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ of Paper 10 describes the UV screening sector, not the dark-energy action.) The power-law sector presented in this appendix should be regarded as a *motivational framework* for the algebraic structure of $w_0$, $w_a$ at the background level; the perturbation-stable formulation is developed in Paper 7.

## 2.1 The SSEE Action

The SSEE dark-energy sector is described by a k-essence scalar field $\phi$ minimally coupled to gravity and extended by a geometric bulk-viscosity term in the Eckart approximation [Eckart, 1940, Weinberg, 1971]:

$$S = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} + P(X) - \zeta\left(\nabla_\mu u^\mu\right)^2 + \mathcal{L}_m\right], \tag{7}$$

where $X \equiv -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \, \partial_\nu \phi$ is the canonical kinetic term. The fluid four-velocity of the dark-energy component is identified with the normalised gradient of the field,

$$u_\mu = -\frac{\partial_\mu \phi}{\sqrt{2X}}, \tag{8}$$

so that $u_\mu u^\mu = -1$ is satisfied automatically, and the viscous and kinetic sectors remain self-consistently defined by the single degree of freedom $\phi$.

## 2.2 Algebraic Determination of $P(X)$

For a power-law k-essence Lagrangian $P(X) = \mathcal{A} X^n$ the equation of state is *exactly* constant:

$$w_\phi = \frac{P}{2X P_X - P} = \frac{\mathcal{A} X^n}{(2n-1)\mathcal{A} X^n} = \frac{1}{2n-1}. \tag{9}$$

Imposing the SSEE algebraic constraint $w_0 = -T_r/M_v$ uniquely fixes the kinetic exponent without any fitting:

$$\boxed{n = \frac{T_r - M_v}{2\,T_r} = \frac{1 + w_0}{2 w_0} \approx -0.09527,} \tag{10}$$

where $T_r = 3(\varphi + \beta)$ and $M_v = \varphi + \pi + K_v$ are the SSEE structural constants defined in Section 1.2. One verifies immediately that $1/(2n-1) = -T_r/M_v = w_0$ identically.

The overall normalisation is fixed by requiring that the present-day dark-energy density equals the SSEE prediction:

$$\rho_{\mathrm{DE},0} = \frac{3H_0^2}{8\pi G}\left(1 - \Omega_m\right), \qquad 1 - \Omega_m = 0.691119, \tag{11}$$

where $\Omega_m = \omega_m/h^2 = 0.308881$. Earlier versions of this equation carried the saturation $T_r/M_v = 0.839950$ in this slot. That quantity is $|w_0|$, a number belonging to the equation of state, and it does not multiply a critical density; the amplitude error is 21.5%. Since $\rho_{\mathrm{DE},0}$ enters only the overall normalisation $\mathcal{A}$ of $P(X)$ and not any dimensionless prediction, no published observable moves; the correction is made because using a saturation where a density belongs is a misuse of both.[1] with $H_0$ constrained observationally to $67.82^{+0.41}_{-0.41}$ km s$^{-1}$ Mpc$^{-1}$ by the MCMC analysis of Paper 2 ($H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}})$ as prior). Given the energy scale $\rho_{\mathrm{DE},0}$, the amplitude reads

$$\mathcal{A} = \frac{\rho_{\mathrm{DE},0}^{1-n}}{2n-1}, \tag{12}$$

which completes the specification of $P(X)$ with no free parameters beyond $H_0$.

**Remark on the CPL evolution.** The pure k-essence sector (9) delivers a *constant* $w = w_0$. The time variation $w_a$ arises from the viscous source (Section 2.3), and is shown there to emerge algebraically from the scale-factor evolution of the viscosity coefficient.

[1] The two are easy to confuse because $T_r/M_v$ and $1-\Omega_m$ are both dimensionless and both near 0.7–0.8. The discriminator is whether the slot carries a factor of $H$: a density does, an equation of state does not.

## 2.3 Geometric Bulk Viscosity and the Emergence of $w_a$

### 2.3.1 Constant-coupling regime ($w_0$)

The retention constant $\mathrm{KAL}_0 \equiv \beta+\pi \approx 5.521$ governs the dissipative sector. Following the Eckart first-order formalism, the bulk viscosity coefficient is coupled to the instantaneous expansion rate:

$$\zeta_0 = \frac{\mathrm{KAL}_0}{8\pi G}\, H. \tag{13}$$

This delivers a *constant-coupling* viscous pressure $\Pi_0 = -3\zeta_0 H = -3\,\mathrm{KAL}_0 H^2/8\pi G$, which governs the background acceleration (Eq. 25) and fixes the present-day effective equation of state $w_0 = -T_r/M_v$ algebraically (Section 2.2).

**Physical motivation for $\zeta \propto H$.** Bulk viscosity proportional to the Hubble rate arises naturally in kinetic theory when the dominant relaxation time is set by the expansion itself [Weinberg, 1971, Brevik et al., 2004]. Within SSEE the coupling constant is not fitted but predicted geometrically as $\mathrm{KAL}_0 = (\pi + \varphi)/2 + \pi$.

### 2.3.2 Scale-factor evolution and the algebraic determination of $w_a$

The constant-coupling viscosity of the preceding section fixes $w_0$ algebraically but predicts a *time-independent* effective equation of state. DESI DR2 data require a time-varying component; within SSEE this must arise from a scale-factor-dependent extension of the viscosity coefficient.

**Structural requirements on $\zeta(a)$.** Any admissible SSEE viscosity extension $\zeta(a)$ must satisfy three independent constraints:

(i) **De Sitter endpoint:** $\zeta(a) \to 0$ as $a \to 1$, so that no viscous term shifts the present-day equation of state away from the k-essence value $w_0 = -T_r/M_v$.

(ii) **Dimensional consistency:** $[\zeta] = [\rho_{\mathrm{DE}}/H]$, constructed from the available dynamical quantities $\{\rho_{\mathrm{DE}}, H, a\}$.

(iii) **Algebraic closure:** the dimensionless amplitude must be a ratio of SSEE structural constants derived exclusively from $\{\varphi, \pi\}$.

The unique admissible form satisfying (i) and (ii) at leading order in $(1-a)$ — the natural expansion around the de Sitter endpoint — is

$$\zeta(a) = \Gamma\, \frac{(1-a)\,\rho_{\mathrm{DE}}(a)}{H(a)}, \tag{14}$$

where $\Gamma$ is a dimensionless amplitude to be fixed by constraint (iii).

**Algebraic determination of $\Gamma$.** The k-essence sector has already employed the ratio $T_r/M_v$ to fix $|w_0|$. The only independent dimensionless ratio remaining among the SSEE structural constants is $P_{sc}/I_g$, where $P_{sc} = 2\varphi + \pi \approx 6.378$ (Dynamical Evolution Scalar) and $I_g$ (IGNIS) $= \pi + P_{sc} \approx 9.519$ (Disruptive convergence, the $w_a$ denominator). The amplitude follows from the viscous pressure itself. With $\Pi = -3\zeta H$ and $\zeta$ of the form (14), the viscous contribution to the equation of state is $\Pi/\rho_{\mathrm{DE}} = -3\Gamma(1-a)$, which must

reproduce the CPL term $w_a(1-a)$. Hence $\Gamma = -w_a/3$, and with the ratio identified above,

$$\Gamma = \frac{P_{sc}}{3\,I_g} = \frac{P_{sc}}{3(\pi + P_{sc})} = +0.223324961891, \tag{15}$$

yielding the unique SSEE-admissible viscosity:

$$\boxed{\zeta(a) = \frac{2\varphi + \pi}{6(\varphi + \pi)} \frac{(1-a)\,\rho_{\rm DE}(a)}{H}.} \tag{16}$$

Earlier versions set $\Gamma = -P_{sc}/I_g$, that is $\Gamma = w_a$ rather than $-w_a/3$. Propagated through $\Pi = -3\zeta H$ that choice returns $w_a = +2.009925$: wrong in sign and by a factor of three. The *identification* of the ratio $P_{sc}/I_g$ is unaffected — it is Eq. (17) that the construction fixes, and it is unchanged — but the intermediate amplitude was misreported, and with the corrected $\Gamma$ the closure $w_a = -3\Gamma$ holds to $0.00 \times 10^0$.

**Algebraic consistency with CPL.** The CPL parametrisation $w(a) = w_0 + w_a(1-a)$ is the standard model-independent description of slowly-varying dark energy; it serves here as the observational language, not as a physical assumption of SSEE. Adopting CPL and substituting $\zeta(a)$ from Eq. (16) into the SSEE conservation equation (26), one finds algebraic consistency if and only if

$$\boxed{w_a = -\frac{P_{sc}}{I_g} = -\frac{P_{sc}}{\pi + P_{sc}} \approx -0.66997.} \tag{17}$$

This is not a derived prediction in the sense that CPL is deduced from the action; rather, it establishes that *the SSEE viscosity uniquely fixes the CPL coefficient $w_a$ from the algebraic structure of $\{\varphi, \pi\}$*. The quantity $w_a$ is therefore not a free parameter of the model but a computable consequence of the two-axiom system, in the same sense that $w_0$ is fixed by the k-essence exponent $n$. At $a = 1$ the viscous correction vanishes ($\zeta \to 0$), recovering the pure k-essence regime $w = w_0$.

The implied dark-energy density evolution reads:

$$\rho_{\rm DE}(a) = \rho_{\rm DE,0}\, a^{-3(1+w_0+w_a)}\, e^{-3w_a(1-a)}, \tag{18}$$

consistent with the CPL conservation equation

$$\dot\rho_{\rm DE} + 3H\,\rho_{\rm DE}\,(1+w_0) = -3H\,w_a\,(1-a)\,\rho_{\rm DE}, \tag{19}$$

with the effective viscous pressure

$$\Pi(a) = w_a\,(1-a)\,\rho_{\rm DE}(a). \tag{20}$$

At $a \ll 1$ the factor $(1-a) \to 1$ and $\rho_{\rm DE}$ redshifts according to Eq. (18), so $\zeta$ remains finite and well-behaved throughout cosmic history.

**Boundary conditions.** The full viscosity interpolates between two algebraically-fixed regimes:

$$a \to 0\ \text{(early)}: \quad \zeta \to -\frac{2\varphi + \pi}{2(\varphi + \pi)} \frac{\rho_{\rm DE,0}\, a^{-3(1+w_0+w_a)}\, e^{-3w_a}}{H(a)}, \tag{21}$$

$$a \to 1\ \text{(today)}: \quad \zeta \to 0. \tag{22}$$

Both limits are free of divergences and require no new parameters.

The viscous pressure is therefore

$$\Pi(a) = \underbrace{-\frac{3\,\mathrm{KAL}_0\,H^2}{8\pi G}}_{\text{background}} \underbrace{-\,3H\,w_a\,(1-a)\,\rho_{\mathrm{DE}}(a)}_{\text{CPL evolution}}, \tag{23}$$

where the first term is the constant-coupling Eckart pressure that sustains $\ddot{a} > 0$ and the second term generates the running $w(a)$ observed by DESI DR2 [Abdul-Karim et al., 2025].

## 2.4 Modified Friedmann Equations

Applying the principle of least action ($\delta S = 0$) to the FLRW metric $ds^2 = -dt^2 + a^2(t)\,d\mathbf{x}^2$ yields three governing equations for the SSEE background.

**I. Energy equation:**

$$H^2 = \frac{8\pi G}{3}\big(\rho_m + \rho_r + \rho_{\mathrm{SSEE}}\big). \tag{24}$$

**II. Acceleration equation:**

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left[\rho_m + 2\rho_r + \rho_{\mathrm{SSEE}}(1+3w_0) - \frac{9\,\mathrm{KAL}_0\,H^2}{8\pi G}\right]. \tag{25}$$

**III. Conservation equation with dissipative source:**

$$\dot{\rho}_{\mathrm{SSEE}} + 3H\,\rho_{\mathrm{SSEE}}(1+w_0) = \frac{9\,\mathrm{KAL}_0\,H^3}{8\pi G}. \tag{26}$$

**Derivation of the coefficient 9.**

From Eq. (13), the viscous pressure is $\Pi = -3\mathrm{KAL}_0 H^2/8\pi G$. In the Raychaudhuri equation the effective pressure of the SSEE sector enters as $p_{\mathrm{eff}} = w_0\rho_{\mathrm{SSEE}} + \Pi$, so the acceleration term becomes

$$\begin{aligned} -\frac{4\pi G}{3}\big(\rho_{\mathrm{SSEE}} + 3p_{\mathrm{eff}}\big) &= -\frac{4\pi G}{3}[\rho_{\mathrm{SSEE}}(1+3w_0) + 3\Pi] \\ &= -\frac{4\pi G}{3}\left[\rho_{\mathrm{SSEE}}(1+3w_0) - \frac{9\,\mathrm{KAL}_0\,H^2}{8\pi G}\right], \end{aligned} \tag{27}$$

which is exactly Eq. (25). The factor 9 arises from $3 \times |\Pi|/(H^2) = 3 \times 3\mathrm{KAL}_0/8\pi G$; a factor of 3 rather than 9 would correspond to including $\Pi$ only once instead of as the trace of the viscous stress.

Similarly, the conservation law $\nabla_\mu T^{\mu\nu} = 0$ for the effective fluid $(w_0, \Pi)$ gives $\dot{\rho} + 3H(\rho + w_0\rho + \Pi) = 0$, hence $\dot{\rho} + 3H\rho(1+w_0) = 9\mathrm{KAL}_0 H^3/8\pi G$, confirming that Eq. (25) and Eq. (26) are mutually consistent.

## 2.5 Israel–Stewart Viscous Perturbations

The Eckart formulation used in §2.2–2.4 is a first-order theory that violates causality [Hiscock & Lindblom, 1985]: the viscous signal propagates instantaneously. Israel–Stewart (IS) theory [Israel, 1976, Israel & Stewart, 1979, Maartens, 1996] resolves this by promoting $\Pi$ to a dynamical variable with a finite relaxation time $\tau_\Pi$:

$$\tau_\Pi H_0\, \tilde{h}(a)\, \frac{\mathrm{d}\tilde{\Pi}}{\mathrm{d}\ln a} + \tilde{\Pi} = -\,\mathrm{KAL}_0\, \tilde{h}^2(a), \tag{28}$$

where $\tilde{h} = H/H_0$, $\tilde{\Pi} = \Pi\, 8\pi G/H_0^2$, and all densities are normalised to $\rho_{\mathrm{crit},0}$. In the limit $\tau_\Pi \to 0$, Eq. (28) reduces to the Eckart result $\tilde{\Pi} = -\mathrm{KAL}_0 \tilde{h}^2$.

**The relaxation time.** The IS relaxation time is fixed algebraically, from the same parents $(\mathrm{KAL}_0, \Omega)$ that fix the viscosity, differing only in which sovereignty sits in the denominator:

$$\tau_\Pi H_0 \;=\; \frac{\mathrm{KAL}_0\, \Omega}{T_r} \;=\; \frac{\mathrm{KAL}_0}{3\,\Omega_{\mathrm{DE}}} \;\approx\; 2.191, \tag{29}$$

with no free parameters.

**A withdrawn justification (2026-09-06).** Earlier versions of this appendix presented Eq. (29) as *derived* by saturating the Israel–Stewart causality bound written as $\zeta/(\rho_{\mathrm{DE}}\tau_\Pi) \le 1$. That argument is withdrawn: the inertia of a sound wave is the enthalpy $\rho + p$, not $\rho$ [Maartens, 1996, Hiscock & Lindblom, 1985], as the dispersion relation of Paper 5 (Appendix C) shows explicitly. With the correct inertia the same saturation condition returns $\tau_\Pi H_0 = \tilde{\zeta}\,(1 + w_0) = 0.2946$, not 2.191; the agreement with Eq. (29) was an artefact of the missing factor $(1 + w_0)^{-1} = 6.248$. Nothing numerical changes: $\tau_\Pi H_0 = \mathrm{KAL}_0\Omega/T_r$ is algebra, on the same footing as every other structural constant of the framework, and it was already stated in that form. What is withdrawn is the claim that causality *derives* it. The bound itself is satisfied with room to spare: with $\tilde{\zeta}$ normalised to the enthalpy — the normalisation both appendices of Paper 5 actually use — the effective sound speed is $c^2_{s,\mathrm{eff}} = w_0 + \tilde{\zeta}/(\tau_\Pi H_0) = 0$, marginally stable and manifestly subluminal.

**Growth index.** Coupling Eq. (28) to the conservation equation (26) and the linear-growth ODE

$$\delta'' + \left[2 + \frac{\mathrm{d}\ln\tilde{h}}{\mathrm{d}\ln a}\right]\delta' = \frac{3}{2}\,\frac{\Omega_m\, a^{-3}}{\tilde{h}^2(a)}\,\delta, \tag{30}$$

we solve the system numerically (`ssee_is_growth.py`) with the IS background $\tilde{h}^2(a) = \Omega_m\, a^{-3} + \rho_{\mathrm{DE}}(a)$, with the *total* matter density $\Omega_m = \omega_m/h^2 = 0.308881$. Both slots previously carried $\Omega_{m,\mathrm{dyn}} = 0.160050$, which is $s_m = 1 + w_0$ — a number of the equation of state, not a density. Re-running the integration with the correct density (`src/p05_is/ssee_is_growth.py`) moves the growth index by $2 \times 10^{-4}$: $\gamma$ is set by $w(a)$, not by $\Omega_m$, so the value below is unaffected. What the substitution *does* change is $S_8$, reported at the end of this paragraph. The growth rate $f \equiv \mathrm{d}\ln\delta/\mathrm{d}\ln a \approx \Omega_m(a)^\gamma$ is well described by a power law over $0.1 \le a \le 1$ with

$$\gamma_{\mathrm{bg}} = 0.658 \pm 0.002 \quad \text{(essentially independent of } \tau_\Pi H_0\text{)}. \tag{31}$$

This is the CDM growth index on the IS background, i.e. Eq. (30) without viscous perturbation feedback in the $\delta$ equation. The full IS-coupled perturbation theory of Paper 5, which includes viscous feedback in the growth equation, yields $\gamma_{\rm IS} = 0.5504 \pm 0.0003$ — consistent with $\gamma_{\Lambda\rm CDM} \approx 0.55$ and lower than the background-only estimate here; $\gamma_{\rm IS}$ is the growth index the suite uses (Paper 3 §5.4), and the target for Stage-IV weak-lensing surveys (LSST, Euclid). An early dimensional-analysis value $\gamma = \varphi^{-1} = 0.618034$ was never derived from the growth equation and is superseded. Note that the direct IS background growth-factor integral (this Appendix) yields $S_8 = 0.823$[2], Because $\sigma_8$ is normalised to Planck 2018, this $S_8$ is Planck's amplitude rescaled by $\Omega_m$: its closeness to the Planck value ($0.832 \pm 0.013$) holds by construction and is not a test. Paper 5's full IS-coupled perturbation theory (including viscous feedback in $\delta$), with $\Omega_{m,\rm CMB} = 0.308881$ in the Poisson source, yields $\sigma_8 = 0.8136$ and an IS linear-growth $S_8 = 0.8256$ at the *single-sector linear* level ($2.7\sigma$ KiDS; the conservative CLASS amplitude ceiling is $S_8 = 0.827$, $2.7\sigma$). These figures hold $A_s$ fixed at the Planck value, which imports the Planck–KiDS tension; $A_s$ is a free parameter of the model. Fitting it to the raw KiDS-1000 $\xi_\pm$ data with a single matter sector (Paper 6) gives $\sigma_8 = 0.7449 \pm 0.0186$, $S_8 = 0.7559 \pm 0.0189$[3] (the $\Lambda$CDM control on the same raw data through the same pipeline gives $0.7571 \pm 0.0194$) — $0.10\sigma$ from KiDS-1000. The two-sector $\varphi$-DM construction previously invoked here is retracted.

**$S_8$ constraint.** Using the IS growth factor $D_{\rm IS}(a)$ and the CMB-sector matter density $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$ ($\omega_m$-direct, Paper 3; the earlier MIRA mapping 0.3199 is superseded),

$$S_8^{\rm IS,\,ceiling} \equiv \sigma_{8,\rm IS}\sqrt{\frac{\Omega_{m,\rm CMB}}{0.3}} \approx 0.827, \tag{32}$$

the all-cold single-sector ceiling, $0.4\sigma$ below the Planck-2018 value $0.832 \pm 0.013$ [Planck Collaboration VI, 2020]. Substituting 0.160050 into the Poisson source instead would give a far lower amplitude, but that substitution is not a choice the model ever had: 0.160050 is $1 + w_0$, minus the dark-energy equation of state, and an equation-of-state parameter enters the Friedmann equations through the *pressure* term, never as a matter density. There is one matter density, $\Omega_m = \omega_m/h^2 = 0.308881$, and it enters every observable that requires one — $E(z)$, $r_d$, the acoustic scale, the Poisson source of linear growth, and lensing alike (Paper 1 §1.4). The full two-sector resolution once developed in Paper 6 is retracted there; with the primordial amplitude free, a single sector matches the raw shear data.

[2]Corrected 2026-09-26. The previously quoted $S_8 = 0.837$ evaluated $\sigma_8\sqrt{\Omega_m/0.3}$ at $\Omega_m = \Omega_{m,\rm dyn} \times$ MIRA $= 0.3199$; the MIRA matter factor was withdrawn on 2026-06-18 when OP-8 closed and $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$ became the derived quantity. Re-running the same integral with the canonical density gives 0.823 at an unchanged $\sigma_8 = 0.811$, the Planck 2018 value that normalises the growth-factor proxy. The growth index is untouched by either substitution.

[3]Updated 2026-09-20. Against the later KiDS-Legacy release (357 raw $\xi_\pm$ points; Wright et al. 2025, arXiv:2503.19441 — published sixteen months before this analysis, over which no temporal priority is claimed) the amplitude the shear asks for lies $0.49\sigma$ from the one the CMB fixes under the *same* algebraic background, so $A_s$ need not be free either: fixed to that value the growth sector carries *no* free cosmological parameter and predicts $S_8 = 0.8273$ against $0.8265 \pm 0.0176$ measured ($0.04\sigma$). The three $\chi^2$ compared span 1.0 over 357 points, so the shear does *not* discriminate between the models; what separates them is the parameter count.

## 2.6 Physical Regimes

**Late universe ($z \lesssim 2$, DESI DR2).** At low redshift $H \approx H_0$, so the viscous source $\propto \mathrm{KAL}_0 H^3/8\pi G$ is small compared to $\rho_{\mathrm{SSEE}}$. The acceleration equation is dominated by $\rho_{\mathrm{SSEE}}(1+3w_0) < 0$ (since $w_0 \approx -0.840 < -1/3$), driving $\ddot{a} > 0$ without dark matter. This regime is the one tested against DESI DR2 in Paper 2, where the algebraic prediction $(w_0, w_a)$ matches the data at $0.24\sigma$ (Pantheon+; 0.2–$1.8\sigma$ across SN compilations).

**Early universe (CMB epoch, $z \sim 10^3$).** At recombination $H \gg H_0$, the geometric retention term $9\,\mathrm{KAL}_0 H^3/8\pi G$ acts as a dissipative source in Eq. (26), injecting energy into the dark-energy fluid and modifying its effective equation of state. The matter density that sets the CMB acoustic scale is the $\omega_m$-direct value $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$ (Paper 3), built from the algebraic pieces $\omega_b$, $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ and $\omega_\nu$, with $\phi$-DM already non-relativistic and contributing to the matter budget at recombination. The earlier mapping of the dynamical $s_m = 0.160050$ to the CMB scale through a MIRA factor $\approx 1.999$ is superseded: there is no matter factor (OP-8 closed; Paper 1 §1.4).

## 2.7 Inflationary Embedding: $\alpha$-Attractor with $\alpha = \varphi^4/3$

The SSEE algebraic predictions for the primordial scalar spectral index $n_s = 1 - \varphi^{-7}$ (Paper 1, §1.1) and the tensor-to-scalar ratio $r = \varphi^{-10}$ (Paper 4) belong simultaneously to a well-defined class of inflationary models [Guth, 1981, Linde, 1982]: the *$\alpha$-attractors* of Kallosh & Linde [2013].

**$\alpha$-attractor predictions.** In the universal leading-order limit, $\alpha$-attractor models predict

$$n_s = 1 - \frac{2}{N}, \qquad r = \frac{12\alpha}{N^2}, \tag{33}$$

where $N$ is the number of inflationary e-folds and $\alpha > 0$ parameterises the inverse curvature of the scalar field's Kähler target manifold [Kallosh & Linde, 2013, Ferrara et al., 2013]. Setting $\alpha = 1$ recovers Starobinsky $R^2$ inflation [Starobinsky, 1980] exactly.

**SSEE consistency check.** The SSEE axioms pre-commit two *independent* inflationary observables: $n_s = 1-\varphi^{-7}$ (Axiom 1, spectral tilt) and $r = \varphi^{-10}$ (Axiom 2, tensor-to-scalar ratio). Neither is derived from the $\alpha$-attractor framework; they are algebraic predictions from $\{\varphi, \pi\}$ that can be tested for *mutual consistency* within that framework.

From Axiom 1 alone, the standard slow-roll relation $n_s = 1 - 2/N$ gives

$$N = \frac{2}{\varphi^{-7}} = 2\varphi^7 \approx 58.07, \tag{34}$$

which lies squarely in the observationally required window $50 \lesssim N \lesssim 60$ [Planck Collaboration X, 2020].

*Independently*, inserting $N = 2\varphi^7$ and Axiom 2 ($r = \varphi^{-10}$) into the second attractor relation $r = 12\alpha/N^2$:

$$\alpha = \frac{r\,N^2}{12} = \frac{\varphi^{-10}\cdot 4\varphi^{14}}{12} = \frac{4\varphi^4}{12} = \frac{\varphi^4}{3}. \tag{35}$$

This is a *non-trivial internal consistency*: two independently pre-committed predictions simultaneously select a unique $\alpha$-attractor geometry. The identification is algebraically

exact (verified to machine precision; see `ssee_inflation_connection.py`). Using the Fibonacci identity $\varphi^4 = 3\varphi + 2$:

$$\alpha = \frac{3\varphi + 2}{3} = \varphi + \frac{2}{3} \approx 2.285. \tag{36}$$

**Geometric interpretation.** In the $\alpha$-attractor framework the Kähler curvature of the inflaton's target space (a Poincaré disk of radius $\sqrt{3\alpha}$) is $\mathcal{R} = -2/(3\alpha)$. For $\alpha = \varphi^4/3$:

$$\mathcal{R} = -\frac{2}{3 \cdot \varphi^4/3} = -\frac{2}{\varphi^4} = -2\varphi^{-4} \approx -0.292. \tag{37}$$

The inflaton therefore lives on a hyperbolic manifold whose curvature is set by $\varphi$ alone. This gives a *geometric* origin for the appearance of $\varphi$ in inflationary observables: $\varphi$ enters not as a numerical coincidence but as the intrinsic length scale of the Kähler geometry of the SSEE inflaton sector.

**Slow-roll parameters.** The SSEE predictions $(n_s,\, r) = (1-\varphi^{-7},\, \varphi^{-10})$ fix the slow-roll parameters uniquely:

$$\varepsilon = \frac{r}{16} = \frac{\varphi^{-10}}{16}, \tag{38}$$

$$\eta = \frac{n_s - 1 + 6\varepsilon}{2} = \frac{-\varphi^{-7} + 3\varphi^{-10}/8}{2}, \tag{39}$$

$$\frac{\eta}{\varepsilon} = 3 - 8\varphi^3 = -5 - 16\varphi \approx -30.9. \tag{40}$$

The ratio $\eta/\varepsilon = 3-8\varphi^3$ is a pure algebraic identity derived from the SSEE axioms $\{\varphi, \pi\}$; it defines a one-parameter family of inflationary potentials $V(\phi_{\rm inf})$ whose single element consistent with $\alpha$-attractors is $\alpha = \varphi^4/3$.

**Relation to Starobinsky.** Starobinsky ($\alpha = 1$) reproduces $n_s$ correctly for $N = 2\varphi^7$ but gives $r_{\rm Staro} = 3/\varphi^{14} \approx 0.00356$, a factor of $\varphi^4/3 \approx 2.28$ below the SSEE prediction $\varphi^{-10}$. SSEE is therefore *not* Starobinsky inflation; it is a generalised $\alpha$-attractor that reduces to Starobinsky only in the limit $\varphi \to 1$ (i.e. $\alpha \to 1$, the fixed point of the golden-ratio recursion $x = 1 + 1/x$).

## 2.8 Summary and Open Problems

Table 2 summarises the EFT status. The action (7) with kinetic exponent (10) and viscosity (16) constitutes a self-consistent, zero-free-parameter description of the SSEE background evolution. Every constant ($n$, $\mathrm{KAL}_0$, $w_0$, $w_a$, $\Omega_{\rm DE}$) is derived from $\varphi$ and $\pi$ alone; only the overall energy scale $H_0$ enters as an observational input. Notably, $w_a = -P_{sc}/I_g$ is *not* postulated but emerges as the unique algebraic amplitude of the viscosity's scale-factor evolution (Section 2.3.2). The gradient instability of the power-law IR sector ($c_s^2 < 0$) is resolved by the two-term ghost-condensate action $K(X) = c_1 X + c_2 X^2$ (Paper 7, $c_s^2 = 0.021284$); the present section focuses on the background algebraic structure.

Four aspects require further development:

1. **Growth index.** The background IS growth index $\gamma_{\rm bg} = 0.658$ (§2.5) omits viscous feedback on $\delta$; the full IS-coupled result (Paper 5) gives $\gamma_{\rm IS} = 0.5504$, close to ΛCDM, and is the value testable by LSST/Euclid Stage-IV weak lensing. The growth index is set by $w(a)$; no mechanism selects $\gamma = \varphi^{-1}$, and that early estimate is superseded.

2. **Full perturbation spectrum.** The IS growth ODE (30) governs sub-Hubble dark-matter perturbations [Mukhanov et al., 1992]; the matching to CMB scalar transfer functions and the tensor-mode propagation in the IS background require a complete Boltzmann implementation.

3. **Radiative stability.** The EFT cutoff scale $\Lambda_{\rm UV}$ and the quantum stability of the $n < 0$ power-law sector are not yet established.

4. **Inflation–dark energy unification.** The $\alpha$-attractor sector (§2.7) and the k-essence dark-energy sector (§2.2) both originate from $\varphi$, suggesting a single unified action. Constructing the quintessential inflation potential $V(\phi)$ that reduces to $P(X) = \mathcal{A}X^n$ at late times and to an $\alpha = \varphi^4/3$ attractor at early times is identified as high-priority future work.

# 3 Dark Energy Evolution

SSEE models dynamical dark energy via the Chevallier–Polarski–Linder (CPL) form [Chevallier & Polarski, 2001, Linder, 2003] $w(a) = w_0 + w_a(1-a)$. Motivated by Eq. (6), the expansion is subject to an effective geometric pressure:

$$H^2 = \frac{8\pi G}{3}(\rho_{\rm bar} + \rho_{\rm DE})\,, \qquad \rho_{\rm DE} = \rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right). \tag{41}$$

The baseline state $w_0$ is set by the static geometric ratio $T_r/M_v$, and the evolution $w_a$ encodes the dynamical departure from saturation driven by $P/I_g$:

$$w_0 = -\frac{T_r}{M_v} \approx -0.840, \qquad w_a = -\frac{P}{I_g} \approx -0.670. \tag{42}$$

**Explicit reduction to the two axioms.** Substituting the algebraic definitions $T_r = \frac{3(3\varphi+\pi)}{2}$, $M_v = 3(\varphi+\pi)$, $P_{sc} = 2\varphi+\pi$, and $K_v = 2(\varphi+\pi)$, both dark-energy observables collapse to closed forms in $\varphi$ and $\pi$ alone:

$$\boxed{w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi+\pi}{2(\varphi+\pi)} = -0.839950}, \qquad \boxed{w_a = -\frac{P}{I_g} = -\frac{2\varphi+\pi}{2(\varphi+\pi)} = -0.669975}. \tag{43}$$

Each value is shown with its *lineage* first and its reduction in $\varphi, \pi$ second, in that order and never the reduction alone. The two denominators reduce to the same number, $2(\varphi+\pi) = 9.519253$, but they are *different entities*: $M_v = \varphi + \pi + K_v$ is the maximal dimensional invariant, while $I_g = \pi + P$ (IGNIS) is the saturation partner of the dynamical scalar. No numerical check can tell them apart—only the construction can—so the construction is what the equation carries. No fitting procedure is involved: these values are uniquely fixed by the geometry of $\varphi$ and $\pi$. The numerators $3\varphi+\pi$ and $2\varphi+\pi$ reflect the three-

and two-fold contributions of $\varphi$ to the saturation horizon, while the common denominator $2(\varphi + \pi)$ is twice the Stability Metric $\Omega = \varphi + \pi$.

**Asymptotic saturation and phantom crossing.** The values $w_0$ and $w_a$ are fixed algebraically and compared with data through the CPL form. While it yields $w_0 + w_a = -1.510$, we employ it strictly within the low-redshift domain of validity ($z \leq 2.3$) for observational comparisons, noting that asymptotic stability at $a \to \infty$ requires the full non-linear saturation boundary $T_r/M_v$ rather than the linear CPL extrapolation.

The CPL form also predicts a *phantom crossing*: setting $w(a_*) = -1$ gives

$$a_* = 1 + \frac{1 + w_0}{w_a} = 0.761 \quad (z_* = 0.314). \tag{44}$$

For $z > z_*$ the effective equation of state has $w < -1$; for $z < z_*$, $w > -1$, reaching $w_0 = -0.839950$ today. *The scalar field cannot carry this crossing.* For the action of Paper 7, $K(X) = c_1 X + c_2 X^2$, the sound speed is $c_s^2 = (1 + w)/(5 - 3w)$ along the whole family of solutions, so any configuration of the field with $w < -1$ has $c_s^2 < 0$, a gradient instability (verified by direct differentiation of $K$, not from the closed form; log `cruce_fantasma_quien_lo_lleva.json`). The field itself sits at $w_\varphi = w_0 > -1$, where $c_s^2 = 0.021284 > 0$. If the crossing is physical it must be carried by the viscous pressure $\Pi$ rather than by the field. With the enthalpy normalisation of the Israel–Stewart sector (Paper 5), $1 + w_{\rm eff} = (1 + w_0)(1 - c)$ with $c = |\Pi|/(\rho + p)$, and reproducing the CPL form requires $c(a) = -w_a(1 - a)/(1 + w_0) = 4.186\,(1 - a)$: the viscous pressure equals the enthalpy exactly at $z_*$ and exceeds it at all $z > z_*$ ($c = 2.09$ at $z = 1$). Whether that regime lies within the validity of Israel–Stewart theory is the open question of how $w_a$ is produced (OP-24).

DESI DR2 [Abdul-Karim et al., 2025] analyses indicate an evolving preference for dynamical dark energy. The SSEE algebraic point $(w_0, w_a) = (-0.840, -0.670)$ falls within the 68% confidence contours of the combined BAO+CMB+DESY5 datasets (Mahalanobis $\chi^2_{2D} = 0.42$, i.e. $0.24\sigma$ against DESI+CMB+Pantheon+, 0.2–$1.8\sigma$ across SN compilations; quantified in Paper 2 [Almeida Vallejo, 2026]).

# 4 Galaxy Cluster Dynamics

## 4.1 Cluster masses in general relativity

SSEE does not modify gravity: its effective-field-theory functions are $\alpha_T = \alpha_M = \alpha_B = 0$ (Paper 7), so the effective gravitational coupling is Newton's ($\mu = 1$; Paper 8, canonical limit). The mass that rich clusters show beyond their baryons is the cold dark matter the model predicts, with density $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ (§1.3). A virialised cluster therefore weighs the whole matter content, and its dynamical mass follows from the baryonic one by the cosmic matter-to-baryon ratio:

$$M_{\rm dyn} = \frac{\omega_m}{\omega_b}\, M_{\rm bar} = \left(1 + \mathrm{KAL}_0\, n_s + \frac{\omega_\nu}{\omega_b}\right) M_{\rm bar} = 6.364\, M_{\rm bar}, \tag{45}$$

with no free parameter. Here $\mathrm{KAL}_0$ is not an amplifier of gravity: it sets how much cold dark matter accompanies each unit of baryons.

### 4.2 Quantitative cluster test

Against all 46 systems of the cluster compilation of Zhang et al. [2026] (their Tables II and III, without selection), Eq. (45) reproduces the Newtonian dynamical masses at $2.32\sigma$ when the baryons include the stellar remnants of a variable initial mass function (IGIMF; Kroupa & Weidner 2003), and fails ($8.64\sigma$) with the canonical IMF. ΛCDM run through the identical comparison gives $2.41\sigma$ and $8.46\sigma$: the two backgrounds differ in $M_{\rm dyn}/M_{\rm bar}$ by under one per cent, so clusters test the baryon census and the matter-to-baryon ratio, not the background (full analysis in Paper 2, Almeida Vallejo 2026).

### 4.3 Gravitational lensing and the Bullet Cluster

Because the extra mass of clusters is cold dark matter and gravity is unmodified, lensing in SSEE is lensing in general relativity by the total matter: Paper 8 obtains the Einstein radius of A1689 at the observed scale in this canonical limit. The Bullet Cluster (1E 0657-56), where the lensing mass peaks on the galaxies rather than on the collisional X-ray gas [Clowe et al., 2006], then receives the standard explanation: the cold dark matter is collisionless and passes through the collision with the galaxies, while the gas is stopped. SSEE inherits this explanation because it contains the same ingredient; it does not add one of its own. The open question that remains is not where that mass sits but what it is made of: the production mechanism behind $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ (OP-19 in `OPEN_PROBLEMS.md`).

**Open problem: dimensional connection to fundamental scales.** The global rate $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is obtained from the measured local rate, and it coincides with the pure number $3(\varphi+\pi)^2 \approx 67.962$ (see Appendix A, footnote †). That coincidence is a comparison, not a derivation: no theoretical bridge connects $\{\varphi, \pi\}$ to the ratio of kilometre, second, and megaparsec scales. A complete first-principles derivation would require a quantum-gravity argument linking the Planck length ($\ell_{\rm Pl} \approx 1.6\times10^{-35}\,\mathrm{m}$) to the cosmic expansion rate ($1\,\mathrm{Mpc} \approx 3.1\times10^{22}\,\mathrm{m}$), a connection 57 orders of magnitude in scale. This derivation lies outside the scope of the present framework and is identified as an open problem for future theoretical work; it is the reason a measured $H_0$ enters as the single dimensional input of Postulate D (§1.3) rather than a derived quantity. Finding that bridge would explain *why* the measured cascade lands on $3(\varphi+\pi)^2$; no cosmological model has yet derived $H_0$ from first principles.

## 5 Domain of Validity

A persistent risk in multi-sector theoretical models is that statements of validity or falsification are conflated across domains that probe different physical regimes. This section explicitly delineates the domain of validity of each SSEE sector and the dataset used to validate it. This delineation applies uniformly to Papers 1–4.

**Dynamic sector (low redshift, $z \lesssim 2.3$).** Governed by $w_0 = -0.839950$, $w_a = -0.669975$ ($s_m = 1 + w_0 = 0.160050$ is an EoS-sector identity, not a density). Validated against: DESI DR2 BAO (13 measurements, $0.295 \le z \le 2.33$), galaxy-cluster mass data (IGIMF baseline, Paper 2). *Result*: $\Delta\mathrm{BIC} = -7.25$ favouring SSEE (Paper 2), $0.24\sigma$ from the official DESI DR2 $w_0w_a$CDM posterior (Pantheon+; 0.2–$1.8\sigma$ across SN compilations) in the $w_0$–$w_a$ plane. With the total gravitating matter density in the background geometry the Cosmic-Chronometer check (32 points) gives

$\chi^2_r(H(z)) = 0.472$, comparable to ΛCDM (0.461); the earlier 1.861 was an artefact of the cold sector 0.160050 in $E(z)$ (Paper 2, § "Posterior predictive check: $H(z)$").

**Observational sector (high redshift, CMB).** Governed by $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$, where $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$ is the algebraic physical matter density (no matter factor; Postulate M retired, OP-8 closed) (Structural Constant Dictionary, Zenodo 10.5281/zenodo.20684908). Validated against: Planck 2018 TT+TE+EE+lensing via CAMB; `plik_lite` TTTEEE likelihood via Cobaya (Paper 3). *Result*: $\chi^2_r = 1.042$ (TT, 1971 multipoles); the canonical $\omega_m$-direct CMB fit gives $\Delta\mathrm{BIC} = -33.83$ (full `plik` MCMC, $N = 2354$) at the global rate $H_0^{\mathrm{glob}} = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ ($k = 2$ vs $k = 6$), with the `plik_lite` + low-$\ell$ point estimate at $-26.03$ ($\chi^2_{\mathrm{SSEE}} = 1003.586$). The earlier legacy Cobaya scan gave $\Delta\mathrm{BIC} = -32.2$ at the superseded MIRA anchor $H_0 = 67.037$. *Claim*: SSEE is *not falsified* by the Planck 2018 CMB spectra. This claim is sector-specific: it refers to the observational sector ($\omega_m$-direct, no matter factor; OP-8 closed), the model's single matter density.

**Algebraic sector (purely geometric).** All constants fixed from $\{\varphi, \pi\}$ with zero data input. Cross-validated against: Planck 2018 ($n_s$, $\Omega_b h^2$, $H_0$), LiteBIRD forecast ($r$), BBN ($\sum m_\nu$), Paper 4. *Result*: most Type A predictions in Appendix B, Table 5 are within $2\sigma$ of published observational values (see Paper 4 for full compilation). The baryon density $\Omega_b h^2 = (\pi-\varphi)/[3(\varphi+\pi)^2] = 0.022418$ ($0.32\sigma$ from Planck 2018, $0.02237 \pm 0.00015$) is classified as a *retrodiction*: its structural algebraic form was identified after the data, superseding the preliminary placeholder $3(\pi-\varphi)/200$ (documented in Table 5 and `AUDIT.md`). *Claim*: the algebraic sector is *not falsified* by any currently available dataset at the $> 3\sigma$ level. A future falsification would require $n_s \neq 1 - \varphi^{-7}$ at $> 3\sigma$ from Planck 2018, or $r \neq \varphi^{-10}$ from LiteBIRD.

## 5.1 Unified Model-Comparison Table

Table 3 consolidates every BIC comparison reported across the four manuscripts into a single cross-comparable register. Each row specifies the dataset, $N$, the cosmological background assumed, $k$ for SSEE and ΛCDM, and the physical hypothesis tested. No BIC result outside this table is claimed as a model-selection outcome.

Table 3: Unified BIC comparison register (all four papers). $\Delta\text{BIC} < 0$: SSEE preferred; $> 0$: $\Lambda$CDM preferred. "Fails" denotes a domain-specific tension, not global falsification.

| Dataset | $N$ | Background | $k_{\text{SSEE}}$ | $k_\Lambda$ | $\Delta$BIC | Hypothesis |
|---|---|---|---|---|---|---|
| DESI DR2 + Planck prior | 16 | $\omega_m$-direct | 2 | 3 | $\mathbf{-7.25}$ | Total matter density in geometry; SSEE favoured (CPL $-6.14$; DIC $-6.47$) |
| Full `plik` MCMC | 2354 | $\omega_m$-direct | 2 | 6 | $\mathbf{-33.83}$ | CMB parsimony, **canonical** (off-diag. TTTEEE+lowl+lensing, Paper 3 §b1) |
| `plik_lite` TTTEEE+lowl | 669 | $\omega_m$-direct | 2 | 6 | $-26.03$ | Compressed-nuisance point est., $\{A_s, \tau\}$ fitted (cross-check; $\Omega_m$=0.308881) |
| Planck 2018 diagonal | 5914 | $\omega_m$-direct | 2 | 6 | $-35.0$ | CMB spectra, diagonal cov. (cross-check; $H_0$=67.962) |
| `plik_lite` TTTEEE | 613 | MIRA bg (legacy) | 2 | 6 | $-32.2$ | CMB parsimony; *superseded by the $\omega_m$-direct rows above* |

**Reading guide.** With the total gravitating matter density in the background geometry, the DESI DR2 comparison gives a single figure, $\Delta\text{BIC} = -7.25$ favouring SSEE (the earlier "+206" penalty was an artefact of using the cold sector $1 + w_0 = 0.160050$ as the background density). The canonical CMB result is the full `plik` MCMC row, $\Delta\text{BIC} = -33.83$ ($k = 2$ vs $k = 6$, $N = 2354$, off-diagonal TTTEEE + lowl + lensing), with a best-fit $\chi^2$ statistically indistinguishable from $\Lambda$CDM ($\Delta\chi^2_{\text{BF}} = -2.78$); the preference is parsimony-driven, not a superior fit. The `plik_lite` + low-$\ell$ point estimate ($\Delta\text{BIC} = -26.03$, $\chi^2_{\text{SSEE}} = 1003.586$ vs 1003.596 with $\{A_s, \tau\}$ fitted, $\Lambda$CDM with its own $\sum m_\nu = 0.06$ eV) and the diagonal-covariance spectra ($\Delta\text{BIC} = -35.0$, inflated effective $N$) are consistent $\omega_m$-direct cross-checks. The legacy MIRA-background values ($-32.2$ Cobaya, $-41.0$ aggressive $k$=1) used the superseded $\Omega_m = 0.31993$ mapping and are retired. *No single BIC summarises the entire framework*: each answers a specific physical question stated in the table.

## 5.2 Falsification Manifesto

**Global falsification rule (theory-level).**

**Global Falsification Criterion**

SSEEis falsified **as a whole** if **any single** sector criterion in Table 4 is met at the stated significance threshold. No parameter re-adjustment is possible because all sectors derive from the same two axioms $\{\varphi, \pi\}$: a failure in one sector cannot be repaired without modifying the axioms, which would change the predictions in all other sectors simultaneously.

**Corollary:** the sector-specific structure (dynamic / observational / algebraic) exists to delineate which datasets test which part of the framework, not to create rescue mechanisms. If DESI DR5+ excludes $w_0 = -0.839950$ at $> 3\sigma$, SSEE is rejected; the observational or algebraic sectors do not save it.

To ensure the framework cannot absorb discrepancies via sector-switching, we also state sector-specific criteria below. These apply uniformly across all four papers and cannot be re-interpreted post-hoc.

**SSEE Falsification Criteria**

**Algebraic sector** is falsified if:

- $n_s \neq 1 - \varphi^{-7}$ at $> 3\sigma$ from Planck 2018; or
- $r \neq \varphi^{-10}$ at $> 2\sigma$ from LiteBIRD ($\sim$2032).

**Dynamic sector** is falsified if:

- DESI DR5+ posterior excludes $(-0.840, -0.670)$ at $> 3\sigma$; or
- $\chi^2_r(H(z)) > 2.5$ with full systematic covariance and the background matter density $\Omega_m = 0.308881$ (the geometry takes the *total* matter density; $1 + w_0$ enters as an equation of state, never in $E(z)$ — see §1.4).

**Observational sector** is falsified if:

- Euclid/KiDS/DES require $\Omega_m < 0.25$ or $\Omega_m > 0.36$ at $z < 1$, excluding the single algebraic matter density $\Omega_m = \omega_m/h^2 = 0.308881$; or
- cosmic shear places $S_8$ more than $3\sigma$ from 0.8273, the value the growth sector predicts once the primordial amplitude is *fixed* to what the CMB gives under the same algebraic background — leaving no free cosmological parameter, and so nowhere to absorb the discrepancy (Paper 6; against KiDS-Legacy, $0.8265 \pm 0.0176$ measured, $0.04\sigma$). A weaker earlier form of this criterion put $S_8$ outside $0.7559 \pm 3 \times 0.0189$ with the amplitude *fitted* to the raw KiDS-1000 $\xi_\pm$; that version is superseded, and the falsifier is now strictly harder to survive, since the model no longer gets to fit the amplitude at all. *(The former $k_{\rm fs}$ falsifier is withdrawn with the particle; see §1.4.)*

**What is NOT a falsification:**

- A $\chi^2_r > 1$ under a non-equivalent cosmological background (i.e. evaluating SSEE with $\Lambda$CDM-calibrated priors it does not assume).
- Invoking the withdrawn two-$\Omega_m$ criterion: $\Omega_{m,\rm dyn} = 0.160050$ ($= 1 + w_0$, i.e. $s_m$) was never a matter density, and the free-streaming scale $k_{\rm fs}$ belonged to the retracted $\varphi$-DM sector; neither is a live prediction (see §1.4).

Table 4: Falsification summary: one-sentence decision rule per sector. This table is the binding reference for all four papers. "Tension" = quantified mismatch within a defined domain; "Falsification" = theory-level rejection.

| Sector | Validated by | Falsified by ($> 3\sigma$) | Not a falsification |
|---|---|---|---|
| **Algebraic** | Closed-dictionary algebraic closure; cross-domain Type A table | $n_s \neq 1 - \varphi^{-7}$ (Planck 2018) or $r \neq \varphi^{-10}$ (LiteBIRD) | Any tension arising from unit-convention mismatch |
| **Dynamic** | DESI DR2 $\Delta$BIC $= -7.25$; $w_0, w_a$ within $0.24\sigma$ (Pantheon+) | DESI DR5+ excludes $(-0.840, -0.670)$; or $\chi^2_r(H(z)) > 2.5$ with full cov. | A $\chi^2 > 1$ from $\Lambda$CDM-calibrated priors the model does not assume |
| **Observational** | Planck 2018 $\Delta$BIC $= -33.83$ (full `plik` MCMC, $\omega_m$-direct canonical, $k = 2$); $r_d^{\rm SSEE} = 147.17$ Mpc ($0.32\sigma$) | Euclid/KiDS $\Omega_m < 0.25$ at $z < 1$ | $\chi^2_r = 1.042$ vs $\Lambda$CDM's 1.043 (non-equivalent background) |

**On cross-sector BIC comparisons.** The $\Delta\mathrm{BIC} = -7.25$ (DESI dynamic) and $-33.83$ (CMB $\omega_m$-direct) are not directly comparable (different $N$, backgrounds, and $\Lambda$CDM $k$ counts). The canonical CMB value is the $\omega_m$-direct $\Delta\mathrm{BIC} = -33.83$ (full `plik` MCMC, $k = 2$, $N = 2354$), with the `plik_lite` point estimate at $-26.03$ ($N = 669$, with $\{A_s, \tau\}$ fitted rather than held at their Planck values) and the diagonal cross-check at $-35.0$ ($N = 5914$). The legacy MIRA-background values ($-32.2$ Cobaya, aggressive $k$=1 $-41.0$) used the superseded $\Omega_m = 0.31993$ mapping and are retired (see Paper 3). The authoritative reference is Table 3.


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## Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (https://arxiv.org/abs/2503.14738). Galaxy-cluster dynamical masses are from Zhang et al. [2026] (https://arxiv.org/abs/2602.06082). Analysis scripts and data files are available at https://github.com/mikealmeida1721/SSEE-Cosmologia.

## A Master Derivation Chain: $\varphi, \pi \to$ Observables

Table 5 provides the complete, unambiguous derivation chain from the two axioms ($\varphi$, $\pi$) to every cosmological observable used in Papers 1–4. This table is the authoritative cross-reference for all four manuscripts. Each intermediate constant is labelled as either a *primary register* (derived in one algebraic step from $\varphi$ and $\pi$) or a *secondary register* (derived from primary registers).

* MIRA $= (\varphi + \beta)/2 = (3\varphi + \pi)/4 = 1.998924\ldots$ is the canonical value of the structural constant. Following the 2026 reframe, MIRA serves as the *screening amplitude* (algebraically fixed from $\{\varphi, \pi\}$ via $\beta = (\varphi+\pi)/2$; it enters $f_{\mathrm{screen}} = s_K/(3\,\mathrm{MIRA})$, Paper 9), consistent with its name (Mirror Ratio). It is no longer the sector-mapping operator: after the 2026 $\omega_m$-direct reframe there is *no* matter factor. The CMB matter density is the algebraic physical $\omega_m = \Omega_b h^2 + \Omega_c h^2 + \omega_\nu$, giving $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$; the EoS-sector identity $s_m = 0.160050$ $(= 1 + w_0)$ is not a density (OP-8 closed; the former two-$\Omega_m$ criterion is withdrawn, §1.4). Similarly, $\mathrm{KAL}_0$ is both a derived constant (Level 2) and a dynamical viscosity operator scaling cluster masses.

‡ $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$ is at $0.32\sigma$ from Planck 2018 ($0.02237 \pm 0.00015$). This is a *retrodiction*: the structural form was identified after the data (OP-1, `OPEN_PROBLEMS.md`), superseding the preliminary placeholder $3(\pi - \varphi)/200 = 0.02285$ ($3.2\sigma$), whose factor 200 was not an algebraic $\{\varphi, \pi\}$ quantity. Using the algebraic input $\Omega_b h^2 = 0.022418$, the formula $\mathrm{KAL}_0 \times \Omega_b h^2 \times n_s$ gives $\Omega_c h^2 = 5.521406 \times 0.02242 \times 0.96556 = 0.119514$ ($0.40\sigma$ from Planck $0.1200 \pm 0.0012$).

§ Canonical CMB-sector matter density ($\omega_m$-direct reframe, 2026): $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$ with $\omega_m = \Omega_b h^2 + \Omega_c h^2 + \omega_\nu = 0.14267$ algebraic, at the single global rate $H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}}) \simeq 67.962\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. No matter factor (OP-8 closed). A Planck `plik_lite` TTTEEE + low-$\ell$ evaluation gives $\chi^2_{\mathrm{SSEE}} = 1003.586$ vs $\chi^2_{\Lambda\mathrm{CDM}} = 1003.596$, $\Delta\mathrm{BIC} = -26.03$ (the full `plik` MCMC gives the canonical $-33.83$). Earlier register forms — geometric $(\pi - \varphi)/(\pi + \varphi) = 0.32010$, MIRA mapping $\Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = 0.31993$, and an intermediate factor $\pi/\varphi$ giving 0.31076 — are all superseded by the $\omega_m$-direct construction.

† The **global background rate** shared by every sector of the suite is $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (Postulate D): the measured SH0ES rate is the single dimensionful input, and the global rate is its output. The pure number $3(\varphi+\pi)^2 \approx 67.962$ has no units and is the target the output is compared against, not the input. No derivation of the *absolute* scale from a fundamental energy is claimed (this is the cosmological-constant problem, $H_0/M_{\rm Pl} \sim 10^{-61}$); one measured dimensional input is the identical status $H_0$ holds in ΛCDM.

# B The SSEE Constant Algebra: Named Registers and Construction Laws

The master table (Appendix A) lists each constant in its pure $(\varphi, \pi)$ algebraic form, which is the load-bearing content for all quantitative results. This appendix documents the same objects as a small *named algebra*: a closed set of registers built from $\varphi$ and $\pi$ by two fixed construction laws. The named labels (BIAL, KAL, AURA, ...) are notation, not physical claims — in the same way the quark labels "up"/"down" carry no spatial meaning yet are standard usage. Each name is exactly equal to its pure $(\varphi, \pi)$ expression; the names are retained only because they make the *construction laws* and the shared-value lineages legible, which bare algebra obscures. Each register also carries a construction-level *functional role*; where observation has borne it out, that role is promoted to a validated *model role* — KAL's retention role becoming the retention constant, seen as viscosity in the fluid layer of Paper 5, and MIRA (the "Mirror Ratio") reflection becoming the Hubble screening amplitude of Paper 9. The functional and model roles are the same physics at the construction and validation stages, not two separate claims; a model role is never assigned before the data warrant it.

**The $\beta$ (BIAL) bifurcation.** All non-root nodes branch from the base coupling $\beta = (\varphi+\pi)/2 = \Omega/2 = 2.379813$. Two routes follow, each obeying its own scaling law:

- **Route $\varphi$ (copy law).** $\varphi + \beta = \text{AURA} = (3\varphi + \pi)/2$ scales by integer multiplication and division (dividends and multiples are permitted). This generates the dimensional-ceiling ladder: MIRA = AURA/2 ($\times\frac{1}{2}$), AURA ($\times 1$), DUAL = AURA $\cdot$ 2 ($\times 2$), TRIAL = AURA $\cdot$ 3 ($\times 3$), CUARTAL = AURA $\cdot$ 4 ($\times 4$). Consecutive ceilings differ by exactly one AURA.
- **Route $\pi$ (non-self-addition law).** $\beta + \pi = \text{KAL}_0 = (\varphi + 3\pi)/2$ scales so that each child must *differ* from its parents; it inherits a combination of functions but may not re-add itself. The dynamical-sector scalars ($P_{sc}$, $K_v$, $M_v$) descend by this route.

**Law of non-self-addition in properties.** Two nodes may share a numerical value yet remain distinct entities with different lineage and function: the value is the carrier, the function is the property. The model-relevant instance is MAR $= \Omega + \pi$ versus SOLAR $= \beta + \text{KAL}_0$, both equal to 7.901219 but reached by different parents; this is the node that enters the $\Sigma m_\nu$ chain of Paper 4. Numerical coincidence is therefore structural information, not redundancy.

**Register role → physical role.** A register's role is promoted to a physical role only after the data warrant it, and never the reverse. The canonical example is $\text{KAL}_0$: its preliminary register role (retention/anchor) was refined to the physical role *structural retention* once the Israel–Stewart sector confirmed it (Paper 5). Where no physical mechanism is yet established the model column is left blank rather than forced. The canonical case of a promotion *after* the data is MIRA: its former interpretive "reflection" role (and the matter-factor mechanism once tracked as OP-8) was dissolved by the 2026 $\omega_m$-direct reframe, and its surviving model role is the algebraic screening amplitude $f_{\rm screen} = s_K/(3\,\text{MIRA})$ of Paper 9 (OP-8 closed).

Table 6: The SSEE constant algebra: each named register, its exact pure $(\varphi, \pi)$ value, the construction law that produces it, its construction-level functional role, and the model role that validates it (blank where the physical mechanism is not yet established). The functional role is borne out by the model role wherever the data have spoken (e.g. KAL retention → viscosity; MIRA reflection → screening). The pure $(\varphi, \pi)$ column is load-bearing throughout; the names are equivalent notation.

| Symbol | Pure $(\varphi, \pi)$ | Construction law | Functional role | Model role (validated) | Ceiling |
|---|---|---|---|---|---|
| $\varphi$ | $\varphi$ | root | Intention | Golden ratio; $\alpha$-attractor inflation | — |
| $\pi$ | $\pi$ | root | Structure | Geometric $\pi$ | — |
| $\Omega$ | $\varphi + \pi$ | root combination | Stability | Stability metric (OMEGA) | — |
| $\beta$ | $(\varphi + \pi)/2$ | $\Omega/2$ — bifurcation | Heat / pulse | Base coupling scalar | — |
| AURA | $(3\varphi + \pi)/2$ | $\varphi + \beta$ — copy | Portal / coupling | Disformal coupling (P8); cancels in $H_0$ cascade | $\times 1$ |
| MIRA | $(3\varphi + \pi)/4$ | AURA/2 — copy | Reflection / mirror | Screening amplitude $f_{\text{screen}}$ (P9) | $\times \frac{1}{2}$ |
| DUAL | $3\varphi + \pi$ | AURA $\cdot$ 2 — copy | Duality | — | $\times 2$ |
| TRIAL | $(9\varphi + 3\pi)/2$ | AURA $\cdot$ 3 — copy | Matter / 3D limit | 3D saturation horizon $\to w_0, \Omega_{\text{DE}}$ | $\times 3$ |
| CUARTAL | $6\varphi + 2\pi$ | AURA $\cdot$ 4 — copy | 4D time | — | $\times 4$ |
| $\text{KAL}_0$ | $(\varphi + 3\pi)/2$ | $\beta + \pi$ — non-self-add | Retention / anchor | Structural retention | — |
| $P_{sc}$ | $2\varphi + \pi$ | $\Omega + \varphi$ — non-self-add | Forge / processing | Dynamical evolution scalar $\to w_a$ | — |
| $K_v$ | $2(\varphi + \pi)$ | $\varphi+\pi+\Omega$ — non-self-add | Crystal / order | Structural constraint $\to w_a$ | — |
| $M_v$ | $3(\varphi + \pi)$ | $\varphi + \pi + K_v$ — non-self-add | Maximal / dominant | Maximal dim. invariant $\to w_0, w_a$ | — |
| MAR/SOLAR | $\varphi + 2\pi$ | $\Omega + \pi$ *vs* $\beta + \text{KAL}_0$ | Tide / radiative | Chain node in $\Sigma m_\nu$ (7.901219) | — |

**Naming policy.** Each named register equals its pure $(\varphi, \pi)$ expression exactly; the name is a label for that combination, retained because it tracks which construction lineage a shared value belongs to — e.g. MAR $= \Omega + \pi$ versus SOLAR $= \beta + \text{KAL}_0$, both 7.901219 but distinct registers. This is the same convention by which "up" and "down" label quark flavours without implying spatial direction. All quantitative results in this series use the bare $(\varphi, \pi)$ forms; the named forms are an equivalent, more legible notation for the algebra and its two construction laws.

# C Unified Symbol Glossary

To avoid ambiguity across Papers 1–4, Table 7 provides the canonical definition of every non-standard symbol used in this series.

Table 7: Unified symbol glossary for the SSEEfour-paper series. All symbols are dimensionless unless otherwise stated.

| Symbol | Full name | First defined |
|---|---|---|
| $\varphi$ | Golden ratio $(1+\sqrt{5})/2$ | Paper 1 §1 |
| $\Omega$ | Stability metric, entity OMEGA $(\varphi+\pi)$ | Paper 1 §1 |
| $\beta$ | Base coupling scalar | Paper 1 §1 |
| AURA | Acoustic Unified Resonance Amplitude $(\varphi+\beta=(3\varphi+\pi)/2\approx 3.998)$ | Paper 1 §2 |
| MIRA | Mirror Ratio $(\text{AURA}/2=(3\varphi+\pi)/4\approx 1.9989)$ | Paper 3 §2.4 |
| $\text{KAL}_0$ | Asymptotic structural retention factor $(\beta+\pi)$ | Paper 1 §1 |
| $P_{sc}$ | Dynamical evolution scalar $(\Omega+\varphi)$ | Paper 1 §1 |
| $K_v$ | Structural constraint invariant | Paper 1 §1 |
| $T_r$ | 3D saturation horizon $(3(\varphi+\beta))$ | Paper 1 §1 |
| $M_v$ | Maximal dimensional invariant $(\varphi+\pi+K_v)$ | Paper 1 §1 |
| $s_m$ | EoS-sector identity $1+w_0$ (not a density) | Paper 1 §1 |
| $\Omega_{m,\text{CMB}}$ | CMB-observed matter density ($\omega_m/h^2$, $\omega_m$-direct; no matter factor) | Paper 3 §2.4 |
| $\gamma_{\text{IS}}$ | Israel–Stewart growth index $(0.5504\pm 0.0003)$ | Paper 5 §6.1 |
| $\tau_\Pi H_0$ | IS relaxation time (dimensionless, $\approx 2.191$) | Paper 1 App.A |

# D Result-Type Classification Protocol

A recurring risk in multi-paper series is that results of different epistemic status are compared without adequate labelling. To ensure methodological transparency across the four manuscripts, all numerical results in this series are classified into one of three categories (see also Table 5, column "Type"):

- **Type A — Algebraic:** The value follows solely from $\varphi$ and $\pi$ through the Level 1–3 chain in Table 5. No observational data enters the computation. These results are *prior-free predictions*.
- **Type M — MCMC-validated:** The value is the median of the posterior distribution from the 100-walker `emcee` pipeline described in Paper 2, using DESI DR2 BAO, Planck 2018 priors, and galaxy-cluster mass data. These results reflect the empirical range accessible within the SSEE parameter family.
- **Type P — Prior-dependent:** The value depends on an external dataset anchor (e.g., Planck TT+TE+EE+lensing as a likelihood) that is not internally derived by the SSEE framework. Type P results quantify the consistency of SSEE with standard cosmological datasets; they are not Type A predictions.

The distinction is particularly important for the $\Delta$BIC comparisons in Papers 2 and 3. In Paper 2, the $\Delta\text{BIC}=-7.25$ (dynamic sector) is a Type M result: it reflects the MCMC posterior relative to $\Lambda$CDM under identical priors. In Paper 3, the canonical $\Delta\text{BIC}=-33.83$ ($\omega_m$-direct, full `plik` MCMC, $k=2$; `plik_lite`+low-$\ell$ point estimate $-26.03$; legacy MIRA-background $-32.2$ superseded) is a Type P result: it depends on the Planck `plik` likelihood as an external CMB anchor. These two BIC values are *not directly comparable* because they operate in different model spaces; the distinction is disclosed explicitly in Paper 3 §4.4.

# E Algebraic Prior Space and Derivation Provenance

A legitimate concern in any zero-free-parameter model is whether the algebraic expressions for observables were selected *after* seeing the data (*post-hoc tuning*) or derived from a pre-committed algebraic structure. This appendix documents the derivation provenance of the key SSEE quantities to enable independent assessment.

## Primary register — arithmetic derivation from $\{\varphi, \pi\}$

All seven structural constants of SSEE are derived from the two axioms in at most two layers of arithmetic ($+$, $\times$, integer scale). Four constants ($\Omega$, $\beta$, $K_v$, $M_v$) are Level 1 — direct combinations of $\varphi$ and $\pi$; three ($\mathrm{KAL}_0$, $P_{sc}$, $T_r$) are Level 2 — using Level 1 constants ($\Omega$, $\beta$) as intermediates:

| Symbol | Name | Definition | Value | Layer |
|---|---|---|---|---|
| $\Omega$ | OMEGA / Stability Metric | $\varphi + \pi$ | 4.759627 | L1 |
| $\beta$ | BIAL / Base Coupling | $(\varphi + \pi)/2$ | 2.379813 | L1 |
| $K_v$ | KRYSTOS$_V$ / Structural Constraint | $\varphi + \pi + \Omega$ | 9.519253 | L1 |
| $M_v$ | ATLAS / Maximal Invariant | $\varphi + \pi + K_v$ | 14.278880 | L1 |
| $\mathrm{KAL}_0$ | Structural Retention | $\beta + \pi$ | 5.521406 | L2 |
| $P_{sc}$ | PYROS / Dynamical Scalar | $\Omega + \varphi$ | 6.377661 | L2 |
| $T_r$ | TRIAL / Saturation Horizon | $3(\varphi + \beta)$ | 11.993542 | L2 |

No constants were introduced *ad hoc*: all seven emerge from systematic enumeration of non-redundant, monotonically ordered arithmetic combinations of $\{\varphi, \pi\}$ up to two layers. The *closure criterion* is exhaustiveness within this family: $M_v = 3(\varphi + \pi)$ is the maximal element at Layer 1, and no additional Layer 2 combination yields a new structurally independent register. Products and integer powers of $\{\varphi, \pi\}$ are excluded from the primary register; they would introduce additional structure beyond the stated axioms.

## AURA and MIRA — derivation from pre-existing constants

**AURA** was defined in the original SSEE-Unificado framework as

$$\mathrm{AURA} = \mathrm{BIAL} + \varphi = \beta + \varphi = \frac{\pi + \varphi}{2} + \varphi = \frac{3\varphi + \pi}{2} \approx 3.9978.$$

**MIRA** follows immediately as MIRA = AURA/2:

$$\mathrm{MIRA} = \frac{\beta + \varphi}{2} = \frac{3\varphi + \pi}{4} \approx 1.9989.$$

Both were defined algebraically before any comparison with Planck 2018 data. MIRA's value 1.9989 is fixed by the algebraic construction [Almeida Vallejo, 2026d], independently of any Planck measurement. Under the 2026 $\omega_m$-direct reframe MIRA is the *screening amplitude* (entering $f_{\mathrm{screen}} = s_K/(3\,\mathrm{MIRA})$, Paper 9), consistent with its name (Mirror Ratio). There is no longer any CMB/dynamical matter ratio: the CMB matter density is the algebraic physical $\omega_m$, giving $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$; the EoS-sector identity $s_m = 0.160050$ ($= 1 + w_0$) is not a density (OP-8 closed; the former two-$\Omega_m$ criterion is withdrawn, §1.4). The earlier near-equality $\mathrm{MIRA} \approx \Omega_{m,\mathrm{CMB}}/s_m \approx 2$ (and the intermediate factor $\pi/\varphi$) are retired as the matter mapping; we make no claim of temporal priority over Planck 2018.

## Five exact consistency identities

The following identities hold exactly in $\{\varphi, \pi\}$ algebra and are verified to machine precision ($< 10^{-16}$ relative error):

$$w_0 = -\frac{\mathrm{AURA}}{\Omega} = -\frac{T_r}{M_v} \tag{46}$$

$$s_m = \frac{\pi - \varphi}{2(\varphi + \pi)} = 1 + w_0 \tag{47}$$

$$\mathrm{MIRA} = |w_0| \cdot \beta \tag{48}$$

$$\mathrm{KAL}_0\big|_{\varphi \leftrightarrow \pi} = \mathrm{AURA} \tag{49}$$

$$\mathrm{AURA}\big|_{\varphi \leftrightarrow \pi} = \mathrm{KAL}_0 \tag{50}$$

Identity (49)–(50) is the deepest structural result: under the discrete exchange $\varphi \leftrightarrow \pi$, the scalar-kinetic constant $\mathrm{KAL}_0 = (\varphi + 3\pi)/2$ maps *exactly* to the photon-sector constant $\mathrm{AURA} = (3\varphi + \pi)/2$, and vice versa. The duality (49)–(50) is an exact algebraic statement and stands on its own; the dark-energy action of Paper 7 contains no coupling of the scalar to matter. The open question of whether this discrete symmetry is an exact symmetry of the full $P(X, \phi)$ action is catalogued as OP-7 (`OPEN_PROBLEMS.md`).

## Spectral index $n_s = 1 - \varphi^{-7}$

The SSEE algebraic commitment is $n_s = 1 - \varphi^{-7}$, fixed by the register structure independently of any inflationary framework (Structural Constant Dictionary, Zenodo 10.5281/zenodo.20684908); we make no claim of temporal priority over the public Planck releases, which predate this work. It is not derived from the $\alpha$-attractor class [Starobinsky, 1980, Kallosh & Linde, 2013, Ferrara et al., 2013]; rather, compatibility with that class is established *a posteriori* in App. A §A.5: the two independent SSEE axioms ($n_s = 1-\varphi^{-7}$ and $r = \varphi^{-10}$) determine $N = 2\varphi^7 \approx 58.07$ and $\alpha = \varphi^4/3$ exactly as a consistency check, not as a derivation. The Planck 2018 value [Planck Collaboration X, 2020] $n_s = 0.9649 \pm 0.0042$ is $0.16\sigma$ from the SSEE algebraic value; this comparison was made after the axiom was committed.

## Baryon density $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2]$

The structural expression $(\pi-\varphi)/[3(\varphi+\pi)^2]$ gives 0.02242, at $0.32\sigma$ from Planck 2018 ($0.02237 \pm 0.00015$). Both sides are pure numbers, as they must be: $\Omega_b h^2$ is dimensionless and so is the ratio on the right. The denominator coincides numerically with the global rate read in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $H_0^{\mathrm{glob}}/\mathrm{km\,s^{-1}\,Mpc^{-1}} \simeq 3(\varphi+\pi)^2$, but we do *not* write the relation as $(\pi-\varphi)/H_0^{\mathrm{glob}}$: that would divide a dimensionless number by a rate, and would make the identity depend on the unit chosen for $H_0$. The content is the pure-number relation, not a division by a physical Hubble rate. This form is classified as a *retrodiction*: it was identified after the data (OP-1, `OPEN_PROBLEMS.md`), replacing the preliminary placeholder $3(\pi-\varphi)/200 = 0.02285$ ($3.2\sigma$), in which the factor 200 was not an algebraic $\{\varphi, \pi\}$ quantity. No parameter is fitted: the denominator $3(\varphi+\pi)^2$ is the same pure number the measured Hubble cascade is compared against.

Table 2: EFT status summary for SSEE.

| Element | Status | Derivation |
|---|---|---|
| Action form $S$ | ✓Complete | Standard GR + k-essence + Eckart |
| $u_\mu(\phi)$ coupling | ✓Complete | $u_\mu = -\partial_\mu\phi/\sqrt{2X}$ |
| Kinetic exponent $n$ | ✓Algebraic | $n = (T_r - M_v)/2T_r$ from $\varphi, \pi$ |
| Equation of state $w_0$ | ✓Algebraic | $w_0 = 1/(2n-1) = -T_r/M_v$ |
| Viscosity coupling $\zeta$ | ✓Algebraic | $\zeta = \mathrm{KAL}_0 H/8\pi G$ |
| Friedmann eqs. I–III | ✓Consistent | Factor 9 verified in §2.4 |
| $w_a$ from EFT | ✓Algebraic | $\zeta(a) \propto (1-a)\rho_{\mathrm{DE}}/H$; see §2.3.2 |
| IS relaxation time $\tau_{\mathrm{II}}$ | ✓Algebraic | $\tau_{\mathrm{II}} H_0 = \mathrm{KAL}_0/(3\lvert w_0\rvert) \approx 2.191$ |
| Gradient stability | ○ IR ansatz unstable | Power-law $P(X) \propto X^n$: $c_s^2 \approx -0.84$ (resolved in Paper 7) |
| Dark-energy action stability | ✓Verified | Paper 7: $K(X) = c_1 X + c_2 X^2$, ghost condensate with $c_s^2 = 0.021284$; no ghost, no gradient instability. (The UV screening functional of Paper 10 is not the dark-energy action.) |
| Growth index $\gamma_{\mathrm{bg}}$ | ✓Numerical | $0.658 \pm 0.002$ (IS background, CDM growth eq.; centre and half-range over the $\tau_{\mathrm{II}} H_0$ candidates); superseded by the full IS treatment, $\gamma_{\mathrm{IS}} = 0.5504$ (Paper 5) |
| $S_8$ (CMB sector) | ✓Numerical | 0.823 (IS bkg, $\sigma_8$ normalised to Planck); 0.827 (single-sector ceiling, Paper 5); 0.8273 predicted against raw KiDS-Legacy $\xi_\pm$ (single sector, $A_s$ fixed by the CMB, no free cosmological parameter; Paper 6), superseding the earlier fit to raw KiDS-1000 with $A_s$ free, $0.7559 \pm 0.0189$. The two-sector 0.758 is retracted |
| $\alpha$-attractor ($\alpha = \varphi^4/3$) | ✓Algebraic | Eq. (35); exact, $N = 2\varphi^7$ |
| Kähler curvature $\mathcal{R} = -2\varphi^{-4}$ | ✓Algebraic | Eq. (37) |
| Non-linear perturbation spectrum | ○ Open | N-body growth and Stage-IV weak lensing (the full Planck `plik` likelihood, with its covariance, is done: Paper 3) |
| Inflation–DE unification | ○ Open | Quintessential inflation no- |

Table 5: Master derivation chain for the SSEEframework. Result types: **A** = pure algebraic (no data input), **M** = MCMC-validated (data-constrained posterior), **P** = prior-dependent (requires external dataset anchor). $\varphi = (1+\sqrt{5})/2 \approx 1.61803$; $\pi \approx 3.14159$.

| Symbol | Definition | Value | Observable link | Type |
|---|---|---|---|---|
| *Level 0 — Axioms* | | | | |
| $\varphi$ | $(1+\sqrt{5})/2$ | 1.618034 | All | A |
| $\pi$ | Archimedes constant | 3.141593 | All | A |
| *Level 1 — Primary Registers* | | | | |
| $\Omega$ | $\varphi + \pi$ | 4.759627 | Stability metric | A |
| $\beta$ | $(\varphi + \pi)/2$ | 2.379813 | Base coupling | A |
| $P_{sc}$ | $\Omega + \varphi$ | 6.377661 | Dyn. evolution | A |
| AURA | $\varphi + \beta$ | 3.997847 | Resonance limit | A |
| *Level 2 — Secondary Registers* | | | | |
| $\mathrm{KAL}_0$ | $\beta + \pi$ | 5.521406 | Viscosity; $M_{\mathrm{cluster}}$ | A |
| $K_v$ | $\varphi + \pi + \Omega$ | 9.519253 | Struct. constraint (entity $\mathrm{KRYSTOS}_V$) | A |
| $I_g$ | $\pi + P_{sc}$ | 9.519253 | $w_a$ denominator (entity IGNIS; same value as $K_v$, distinct entity) | A |
| $T_r$ | $3(\varphi + \beta)$ | 11.993542 | Saturation horizon | A |
| MIRA | $\mathrm{AURA}/2 = (\varphi+\beta)/2$ | 1.998924 | Screening amplitude (P9) | A* |
| $M_v$ | $\varphi + \pi + K_v$ | 14.278880 | Max. dim. invariant | A |
| *Level 3 — Cosmological Observables* | | | | |
| $w_0$ | $-T_r/M_v$ | $-0.839950$ | DESI DR2 | A |
| $w_a$ | $-P_{sc}/I_g$ | $-0.669975$ | DESI DR2 | A |
| $H_0^{\mathrm{glob}}$ | $H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}})$ | 67.962 km/s/Mpc | target $3(\varphi+\pi)^2$ | P† |
| $H_0^{\mathrm{MCMC}}$ | DESI DR2 BAO, prior $H_0^{\mathrm{glob}}$, CAMB $r_d$ (Paper 2) | $67.82 \pm 0.41$ km/s/Mpc | DESI DR2 | M |
| $s_m$ | $(\pi - \varphi)/[2(\varphi+\pi)]$ | 0.160050 | EoS sector; $1 + w_0$ exact | A |
| $\Omega_m^{\mathrm{CMB}}$ | $\omega_m/h^2 = (\Omega_b h^2 + \Omega_c h^2 + \omega_\nu)/h^2$ | 0.308881§ | Planck 2018 | A |
| $n_s$ | $1 - \varphi^{-7}$ | 0.965558 | Planck 2018 | A |
| $\Omega_b h^2$ | $(\pi - \varphi)/[3(\varphi+\pi)^2]$ | 0.022418 ($0.32\sigma$) | Planck 2018 | A‡ |
| $\Omega_c h^2$ | $\mathrm{KAL}_0 \times \Omega_b h^2 \times n_s$ | 0.119514 ($0.4\sigma$) | Planck 2018 | A |
| $r$ | $\varphi^{-10}$ | 0.008131 | LiteBIRD 2032 | A |
| $\sum m_\nu^{\mathrm{active}}$ | $\mathcal{R}_2\, \Omega_b h^2 \cdot 93.14/(\tau_\Pi H_0)$ (Paper 4) | 0.06849 eV | Planck+BBN | A |

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